Algebra II

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1 Category Theory

Suggested papers:

1. Eilenberg and MacLane: *General Theory of Natural Equivalences*
2. B. Mazur: *When is one thing equal to some other thing?*

The first object we study in this course is a *category*. While quite abstract, this object allows a common language to use in mathematics to describe equivalences (e.g. natural transformations vis à vis "unnatural" transformations) and relations between mathematical structures (functors).

Definition 1. A *category* $\mathcal{C}$ consists of a class of objects $\mathrm{ob}(\mathcal{C})$ and for every pair of objects a set[1] $\mathrm{Hom}_{\mathcal{C}}(A, B)$. In the case the pair $(A, B) \neq (C, D)$ these morphisms are disjoint sets. Finally, for every triple $A, B, C \in \mathrm{ob}(\mathcal{C})$ we have a map of sets $\mathrm{Hom}(A, B) \times \mathrm{Hom}(B, C) \to \mathrm{Hom}(A, C)$ satisfying the following two properties:

1. For all $A \in \mathrm{ob}(\mathcal{C})$, there is $\mathrm{id}_A \in \mathrm{Hom}(A, A)$ such that for all $B \in \mathrm{ob}(\mathcal{C})$ and $g \in \mathrm{Hom}(A, B)$, $g = g \circ \mathrm{id}_A$ and for $f \in \mathrm{Hom}(B, A)$, $f = \mathrm{id}_A \circ f$.
2. Composition is associative.

I assume here the reader is familiar with common examples of categories, such as <u>Set</u>, <u>Mon</u> (for monoids), <u>Grp</u>, and <u>Ab</u> (for abelian groups).

N.B. The existence of morphisms between objects in a category is not necessarily guaranteed unless one looks at the morphisms of an object to itself, in which case the identity morphism always exists.

We can express statements about morphisms in terms of commutative diagrams. For example, the identity can be expressed saying the following diagram commutes:

$$\begin{tikzcd}
B \arrow[d, "g"] \arrow[dr, "g"] & \\
A \arrow[r, "\mathrm{id}_A"] \arrow[dr, "f"'] & A \arrow[d, "f"] \\
& B
\end{tikzcd}$$

and the associativity of composition can be expressed as the commutivity of

[1] It is not necessary that this is a set; it could very well be a class. With this restriction, we are technically defining a *locally small* category. When $\mathrm{ob}(\mathcal{C})$ is a proper set, we call $\mathcal{C}$ a *small category*.

$$\begin{tikzcd}
 & B \arrow[r, "g"] \arrow[dr, "h\circ g"] & C \arrow[dr, "h"] & \\
A \arrow[ur, "f"] \arrow[urr, "g\circ f"] \arrow[rrr, "h\circ g\circ f"'] & & & D
\end{tikzcd}$$

1.1 Subcategories

Just as with other mathematical machinary, such as sets, groups, or topological spaces, we should have a way of restricting the focus of a category. We do this by way of *subcategories*.

Definition 2. A category $\mathcal{D}$ is a *subcategory* of $\mathcal{C}$, if $\mathrm{ob}(\mathcal{D})$ is a subclass of $\mathrm{ob}(\mathcal{C})$, the morphisms between objects in $\mathcal{D}$ are morphisms between the same objects in $\mathcal{C}$ (i.e. $\mathrm{Hom}_{\mathcal{D}}(A, B) \subset \mathrm{Hom}_{\mathcal{C}}(A, B)$), and the composition law in $\mathcal{D}$ is the one inherited from $\mathcal{C}$.

N.B. In class it was asked whether subcategories also have to inherit the same identity elements; it was stated that it must, but this does not seem immediate from definition

Thus, we see that subclasses are simply restrictions of a cateogry both in the objects and the morphisms; the compositional structure of the arrows are preserved. In the case the subcategory only restricts the objects and not the morphisms, we say $\mathcal{D}$ is a *full* subcategory of $\mathcal{C}$ (i.e. $\mathrm{Hom}_{\mathcal{D}}(A, B) = \mathrm{Hom}_{\mathcal{C}}(A, B)$ whenever the left-hand side is defined).

N.B. The compositional structure of the morphisms of an object to itself should look familiar: it is a monoid. Moreover, from any monoid M, we can create a category $\underline{M}$ with one object whose morphisms give the monoid M.

1.2 Special Morphisms

A special type of morphism in a category is an *isomorphism*. In the case of cateogries one has previously encountered in math, these are exactly the same as isomorphisms in the traditional sense.

Definition 3. We say $f \in \mathrm{Hom}(A, B)$ is an *isomorphism* if there is $g \in \mathrm{Hom}(B, A)$ such that $g \circ f = \mathrm{id}_A$ and $f \circ g = \mathrm{id}_B$.

N.B. Note that by the uniqueness of the identity for an object in a category, the above is equivalent to specifying the morphism has a left and right inverse (which *a priori* may not be the same).

Especially in the case of constructing a category from a monoid (and giving no care as to what the object of said category was), one has the feeling that the most important bits of the cateogry are the morphisms. As we shall see later on the course, this is correct. Indeed, by Yonneda, the morphisms in some sense determine the object.

Following this theme of utilizing the importance of morphisms, one can create a new category from an old by merely reversing the direction of arrows in the category. The result of this action is the opposite category (denoted $\mathcal{C}^{op}$). This category has some use when we define contravariant functors.

Definition 4. A morphism f is called *monic* if it satisfies the left-cancellation law—$f \cdot g_1 = f \cdot g_2 \Rightarrow g_1 = g_2$.

A morphism f is called *epic* if it satisfies the right-cancellation law—$g_1 \cdot f = g_2 \cdot f \Rightarrow g_1 = g_2$.

N.B. In the case of some categories (like Set), the notion of monic and epic exactly corresponds to injectivity and surjectivity. However, as we will see below, this does not always hold (as in Ring).

Here are some easy facts that are proven as soon as one writes them down.

Facts.

1. Epic and monic are preserved under composition.
2. If $g \cdot f$ is monic, so is f.
3. If $g \cdot f$ is epic, so is g.
4. If f has a left inverse, it is monic.
5. If f has a right inverse, it is epic.

We now show an example in which monic and epic indeed correspond to injective and surjective:

Proposition 1. *Fix R a ring. In R-mod, monic and epic corresponds to injective and surjective respectively.*

Proof. The fact injective and surjective imply monic and epic respectively follows by passing through the case in Set.

Suppose that $f : A \to B$ is not injective. Then $\ker f$ is an R-mod, and we have morphisms $i : \ker f \to A$ by inclusion and $z : \ker f \to A$ by the 0-map. It is immediate to see these are different morphisms, as $\ker f \neq 0$. However, $f \cdot i = f \cdot z$, implying f is not monic.

Likewise, if $f : A \to B$ is not surjective, the cokernel $B/f(A)$ is non-zero, and the morphisms $B \to B/f(A)$ by the 0-map and projection are not equal, but their compositions with f are; thus, f is not epic. $\square$

N.B. Note that much of the above proof also applies to Grp, except epic implying monic, as $f(A)$ may not be a group. A bit more care is needed to deal with this case, but it still holds.

In the category Ring the above follows except epic may not imply surjective. The proof is the same, expect to show monic implies injective, one uses the subring of $A \times A$ consisting of pairs (a, b) where $f(a) = f(b)$ and the morphisms of projection onto the first and second coordinate, respectively. The classic example of an non-surjective epimorphism is $\mathbb{Z} \hookrightarrow \mathbb{Q}$. In general, when we have a morphism $R \to T$ where the only new data in $T \setminus R$ consists only of inverses of elements of R we have an epimorphism—the structure of the ring and homomorphism determines what these inverse get sent to.

1.3 Functors and Naturality

"The engineer exists!"

Cogdell

Definition 5. A *covariant functor* F between categories $\mathcal{C}, \mathcal{D}$ consists of a map $F : \mathrm{ob}(\mathcal{C}) \to \mathrm{ob}(\mathcal{D})$ and for every pair of $A, B \in \mathrm{ob}(\mathcal{C})$ a map $F : \hom_{\mathcal{C}}(A, B) \to \hom_{\mathcal{D}}(F(A), F(B))$ satisfying:

1. $F(f \cdot g) = F(f) \cdot F(g)$ whenever $f\cdot$ is defined.
2. $F(\mathrm{id}_A) = \mathrm{id}_{F(A)}$.

This definition is the same as saying

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[dr, "f\cdot g=h"'] & B \arrow[d, "f"] \\ & C \end{tikzcd}$$

implies

$$\begin{tikzcd} F(A) \arrow[r, "F(g)"] \arrow[dr, "F(f)\cdot F(g)=F(h)"'] & B \arrow[d, "F(f)"] \\ & F(C) \end{tikzcd}$$

Definition 6. A *contravariant functor* $F : \mathcal{C} \to \mathcal{D}$ consists of $F : \mathrm{ob}(\mathcal{C}) \to \mathrm{ob}(\mathcal{D})$ and a map on morphisms $F : \hom_{\mathcal{C}}(A, B) \to \hom_{\mathcal{D}}(F(B), F(A))$ such that

1. $F(\mathrm{id}_A) = \mathrm{id}_{F(A)}$.
2. $F(f \cdot g) = F(g) \cdot F(f)$.

This composition law can be written as

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[dr, "f\cdot g=h"'] & B \arrow[d, "f"] \\ & C \end{tikzcd}$$

implies

$$\begin{tikzcd} F(A) & F(B) \arrow[l, "F(g)"'] \\ & F(C) \arrow[u, "F(f)"'] \arrow[ul, "F(h)=F(g)\cdot F(f)"] \end{tikzcd}$$

Example 1. The identity functor $1_{\mathcal{C}} : \mathcal{C} \to \mathcal{C}$.

Example 2. If $\mathcal{D}$ is a subcategory of $\mathcal{C}$, we have an *injection functor* $\mathcal{D} \hookrightarrow \mathcal{C}$

Example 3. A class of functors are *forgetfull functors* which basically forget certain structures in the objects of the category.

For an example, $\underline{Mon} \to \underline{Set}$ is given by $F(M)$ is the underlying set of the monoid M. For $f \in \hom_{\underline{Mon}}(A, B)$, $F(f)$ is just viewed as f as a set map.

However, the functor on the morphisms are largely non-surjective.

Similarly we have forgetful functors
$\underline{Grp} \to \underline{Set}$
$\underline{Ring} \to \underline{Set}$
$\underline{Ring} \to \underline{Ab}$ by forgetting the multiplicative structure,
and $\underline{Ring} \to \underline{Mon}$ by forgetting the additive structure.

Example 4. We have $\underline{Grp} \to \underline{Ab}$ by $F(G) = G/(G, G)$ and $f : G \to H$ is sent to $F(f) : G/(G, G) \to H/(H, H)$ by the induced map.

Example 5. We have $\underline{Ring} \to \underline{Grp}$ by $F(R) = R^{\times} = U(R)$.

Here are some examples of *contravariant functors*:

Example 6. The *opposition* or *dual functor* $\mathcal{C} \to \mathcal{C}^{op}$ is defined by $F(A) = A$ and $F(g) = g$.

N.B. One can compose functors to get another functor.

By composing with the opposition functor, any covariant functor $\mathcal{C} \to \mathcal{D}$ can be made into a contravariant functor $\mathcal{C}^{op} \to \mathcal{D}$.

Example 7. Another duality functor is $D : R - \underline{mod} \to \underline{mod} - R$ by $M \mapsto M^* = \hom(M, R)$.

Note that M^* is naturally a right R-module—we define addition and scalar multiplication by $(f + g)(x) = f(x) + g(x)$ and $(f \cdot r)(x) = f(x) \cdot r$.

The functor on the morphisms acts as follows—For $L \in \hom(M, N)$, $D(L) : \hom(N, R) \to \hom(M, R)$ by $D(L)(\phi) = \phi \circ L$. Sometimes we denote $D(L) = L^*$, and it is the *transpose*.

Definition 7. A functor $F : \mathcal{C} \to \mathcal{D}$ is *faithful* if the map on morphisms is injective.

Example 8. All the forgetful functors are faithful—the morphisms are exactly determined by their actions on the underlying sets.

Definition 8. A functor is *full* if the map on morphisms is surjective.

Example 9. The duality functor $D : R - \underline{mod} \to \underline{mod} - R$ is faithful and full.

Definition 9. We say functors $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ are *inverses* if $G \circ F = 1_{\mathcal{C}}$ and $F \circ G = 1_{\mathcal{D}}$.

Definition 10. If $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{C} \to \mathcal{D}$ are covariant functors, a *natural transformation*[2] $\eta : F \to G$ assigns for every $A \in \mathrm{ob}(\mathcal{C})$ a morphism $\eta_A \in \hom_{\mathcal{D}}(F(A), G(A)$ such that the following diagram commutes:

$$\begin{tikzcd} A \arrow[d, "f"] & F(A) \arrow[r, "\eta_A"] \arrow[d, "F(f)"] & G(A) \arrow[d, "G(f)"] \\ B & F(B) \arrow[r, "\eta_B"] & G(B) \end{tikzcd}$$

This means $G(f) \circ \eta_A = \eta_B \circ F(f)$ for all $f \in \hom_{\mathcal{C}}(A, B)$.

In the case η_A is an isomorphism for all $A \in \mathrm{ob}(\mathcal{C})$, η is called a *natural isomorphism*[3].

In this case, we can define an inverse natural transform $\eta^{-1} : G \to F$ by $(\eta^{-1})_A = \eta_A^{-1}$.

Easily, we can see there is an identity natural transformation for any functor.

We now want to show that there is a natural transformation from the identity functor to the double dual functor.

Example 10. The original example from Eilenberg Mac Lane. We have the contravariant duality functor $D : R - \underline{mod} \to \underline{mod} - R$ by $M \mapsto \hom(M, R)$.

$$\begin{tikzcd} M \arrow[r, "D"] \arrow[d, "L"] & M^* \arrow[r, "D"] & M^{**} \arrow[d, "L^{**}"] \\ N \arrow[r, "D"] & N^* \arrow[r, "D"] \arrow[u, "L^*"] & N^{**} \end{tikzcd}$$

So D^2 is contravariant, $D^2(M) = M^{**}$ Recall the definition of $L^* : g \cdot L = L^*(g)$.

This means the definition of L^{**} is

[2] Also known as a "natty transform"

[3] A.K.A a "natty ice"

$$\begin{tikzcd} N^* \arrow[r, "L^*"] \arrow[dr, "L^{**}(\phi)=\phi\cdot L^*"'] & M^* \arrow[d, "\phi"] \\ & R \end{tikzcd}$$

If $g : N \to R$ (i.e. $g \in N^*$) then $(L^{**}\phi)(g) = \phi(L^*(g)) = \phi(g \cdot L)$.

If $x \in M$ then x definies an element $\eta_M(x) \in M^{**}$ by evaluation—$\eta_M(x)(f) = f(x)$ where $f \in \hom(M, R)$. The claim is that η_M defines a natural transformation $\mathrm{id}_{R-\underline{mod}} \to D^2$.

$$\begin{tikzcd} M \arrow[d, "L"] & M \arrow[r, "\eta_M"] \arrow[d, "L"] & M^{**} = D^2(M) \arrow[d, "L^{**}"] \\ N & N \arrow[r, "\eta_N"] & N^{**} = D^2(N) \end{tikzcd}$$

We have η a natural transform if this diagram commutes for all M, N, L. Let $x \in M$. $L(x) \in N$. Then $\eta_N(L(x))(g) = g(L(x)) = (gL)(x) \in R$. In the other direction if $\phi = \eta_M(x) \in M^{**}$, then

$$L^{**}(\eta_M(x))(g) = \eta_M(x)(L^*g) = (L^*g)(x) = (gL)(x)$$

Therefore the diagram commutes and D^2 is a natural transformation.

N.B. Note that if $R = k$ is a field, then $K - \underline{mod} = K - \underline{vect}$. If moreover we restrict to finite dimensional case, we actually get a natural isomorphism.

Also, the signle dual operation is not natural, as it requires choosing a basis.

Example 11 (Abelianization)**.** $F : \underline{Grp} \to \underline{Ab}$ by $G \mapsto G/(G, G) = G^{ab}$. We can compose this with the injection functor to get $F' : \underline{Grp} \to \underline{Ab} \to \underline{Grp}$. Letting $\nu_G : G \to G^{ab}$ we have the following commuting diagram:

$$\begin{tikzcd} G \arrow[r, "\nu_G"] \arrow[d, "f"] & G^{ab} \arrow[d, "F'(f)"] \\ H \arrow[r, "\nu_H"] & H^{ab} \end{tikzcd}$$

ν then defines a natural transformation.

1.4 Isomorphisms and Equivalences of Categories

Definition 11. We say categories $\mathcal{C}, \mathcal{D}$ are *isomorphic* if we have functors $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ such that $G \circ F = 1_{\mathcal{C}}$ and $F \circ G = 1_{\mathcal{D}}$.

This is a really strong equivalence, and when we do so, we identify the categories.

Example 12. $\underline{Ab}$ is isomorphic to $\mathbb{Z} - \underline{mod}$.

$R - \underline{mod}$ is isomorphic to $\underline{mod} - R^{op}$ wher R^{op} just reverses the multiplication operation.

The more interesting notion of equivalence is *equivalence of categories.*

Definition 12. $\mathcal{C}, \mathcal{D}$ are *equivalent* if $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$ and $G \circ F \cong 1_{\mathcal{C}}$ and $F \circ G \cong 1_{\mathcal{D}}$ where $\cong$ indicates a natural isomorphism.

N.B.
- Isomorphic categories are equivalent.

- Equivalence of cateogries is an equivalence relation. This really says we can compose natural isomorphisms.

Given an isomorphism of categories, then F and G are honest inverses and F determines G. However, this is not true for a simple equivalence—F can vary by isomorphisms on its map of objects. Can one tell, just on level of F whether F determines an equivalence of categories? The answer is yes, as the following proposition gives us.

Proposition 2. *$F : \mathcal{C} \to \mathcal{D}$ is an equivalence of categories if and only if F is fully faithful and essentially surjective*

Proof. If F is an equivalence of categories, $G \circ F \cong 1_{\mathcal{C}}$ then $\hom_{\mathcal{C}}(A, B) \cong \hom_{\mathcal{C}}(GF(A), GF(B))$ So F must by faithful. Similarly $F \circ G \cong 1_{\mathcal{D}}$ $\hom_{\mathcal{D}}(A, B) \cong \hom_{\mathcal{D}}(FG(A), FG(B))$, so F is surjective on the morphisms and full. Finally, if $A' \in \mathrm{ob}(\mathcal{D})$ then $F \circ G \cong 1_{\mathcal{D}}$, so there is an isomorphism $\eta_{A'}$ in $\mathcal{D}$ between $FG(A')$ and A'. This gives that F is essentially surjective (its range is surjective up to isomorphisms in $\mathcal{D}$.)

We construct $G : \mathcal{D} \to \mathcal{C}$ determining an equivalence of categories with F. SKETCH: Use 3 to define $\mathcal{D}$ on objects—for $A' \in \mathrm{ob}(\mathcal{D})$ take $A \in \mathrm{ob}(\mathcal{C})$ with $F(A)$ isomorphic to A'. Define $G(A') = A$. On morphisms, we have F fully faithful, so we can uniquely find inverses. □

$$\begin{tikzcd} A' \arrow[r, "\eta_{A;}"] \arrow[d, "f'"] & F(A) \arrow[d, dotted, "F(f)"] \\ B' \arrow[r, "\eta_{B'}"] & F(B) \end{tikzcd}$$

$\eta_{B'} \circ f' \circ \eta_{A'}^{-1} \in \hom(F(A), F(B))$. Then use fully faithful to say $\exists! f \in \hom(A, B)$ so that the above diagram commutes. Take $G(f') = f$.

One then must check G defines a functor and (F, G) give equivalences.

Example 13. Let R be a ring, $M_n(R)$ $n \times n$ matrices over R. Then $\underline{mod} - R$ and $\underline{mod} - M_n(R)$ are equivalent:

Proof. $F : \underline{mod} - R \to \underline{mod} - M_n(x)$ by M R-module, let $M^{(n)} = M \oplus \cdots \oplus M$. Say $F(M) = M^{(n)}$ and for $f \in \hom(M, N)$ let $F(f) = f^{(n)}$ where $f^{(n)}(m_1, \ldots, m_n) = (f(m_1), \ldots, f(m_n))$. Check this is fully faithful and essentially surjective. □

"I put it on Fridays just because I do" "The answer is no because we have class on that Friday and that would be evil"

1.5 Products and Co-products

> I don't want to say you can't do that,
> but it hurts my head.
>
> *Cogdell*

The main idea is to generalize the idea of the cartesian product of sets or groups to arbitrary sets. However, the classical product is defined in terms of pairs of elements, which we don't have in general cateogry theory. Thus we have to determine a way to do this just using morphisms. Once we do this, dualizing gives the coproduct.

Two important properties of the product: If $G = G_1 \times G_2$ there are 2 projections: $G_1 \times G_2 \to G_1$ and $G_1 \times G_2 \to G_2$ and these are morphisms in the category. If H is a group, and morphisms $H \to G_1$ and $H \to G_2$ then there is a unique $f : H \to G_1 \times G_2$ making the morphisms commute with the projections. This is expressed in the following commutative diagram.

$$\begin{tikzcd}
 & H \arrow[dl, "f_1"'] \arrow[d, dotted, "\exists! f"] \arrow[dr, "f_2"] & \\
G_1 & G \arrow[l, "p_1"] \arrow[r, "p_2"'] & G_2
\end{tikzcd}$$

In groups, f must be defined by $f(h) = (f_1(h), f_2(h))$.

With this notion in hand, we define a general product in a category:

Definition 13. Let $A_1, A_2 \in \mathrm{ob}(\mathcal{C})$. The *product* of A_1, A_2, denoted $A_1 \times A_2$ is a triple (A, p_1, p_2) where $A \in \mathrm{ob}(\mathcal{C})$ and $p_i \in \hom_{\mathcal{C}}(A, A_i)$ satisfying the following universal property: If $B \in \mathrm{ob}(\mathcal{C})$ and $f_i \in \hom_{\mathcal{C}}(B, A_i)$, then there is a unique $f \in \hom(B, A_1 \times A_2)$ such that the following diagram commutes:

$$\begin{tikzcd}
 & B \arrow[dl, "f_1"'] \arrow[d, dotted, "\exists! f"] \arrow[dr, "f_2"] & \\
A_1 & A_1 \times A_2 \arrow[l, "p_1"] \arrow[r, "p_2"'] & A_2
\end{tikzcd}$$

The product $A_1 \times A_2$ can be written as A, $A_1 \times A_2$ or $A_1 \prod A_2$.

N.B. In the definition we used the words "universal mapping property"—this corresponds to the existence of a unique morphism between objects in the category. In particular, this gives products in a category is unique up to isomorphisms (if they exist). A big question that remains then is if products exist in a certain category.

Note this definition can also generalize to arbitrary products (not just finite) in the obvious way.

Definition 14. Let $\{A_\alpha\}_{\alpha \in A}$ be an index set of objects of $\mathcal{C}$. A *product* of $\{A_\alpha\}$ is a tuple consisting of the data $A \in \mathrm{ob}(\mathcal{C})$ and morphisms $p_\alpha : A \to A_\alpha$ satisfying the universal property that if $B \in \mathrm{ob}(\mathcal{C})$ with morphisms $f_\alpha : B \to A_\alpha$ there is a unique morphisms $B \to A$ such that $f_\alpha = p_\alpha \circ f$ for all α.

This is a true extension of the definition, as when $|A| = 2$, this is exactly the product defined above. We write $A = \prod_\alpha A_\alpha$, and if it exists it is unique up to isomorphism.

Example 14.

1. Arbitrary products exist in <u>Sets</u>— Given $\{A_\alpha\}_{\alpha \in I}$. Define $A := \left\{a : I \to \bigcup_{\alpha \in I} A_\alpha \text{ so that } a(\alpha) \in A_\alpha\right\}$. Define $p_\alpha(a) = a(\alpha)$.

   We claim this is a product in <u>Set</u>. Given $(B, \{f_\alpha\})$ where $f_\alpha : B \to A_\alpha \in \hom(B, A_\alpha)$. Then we must have $f : B \to A$ by $f(b) \mapsto a$ where $a(\alpha) = f_\alpha(b)$. This clearly makes everything commute, and it must be this, so it is the unique such morphism $B \to A$.

   A can also be realized as tuples of elements of A_α, and this is the way we have typically viewed Cartesian products

2. In <u>Grp</u>, we form $G = \prod_\alpha G_\alpha$ as defined in algebra class (component-wise multiplication).

3. In <u>Ring</u>, the same idea of the product holds.

The idea of the coproduct is the exact dual of the product.

Definition 15. In a category $\mathcal{C}$, the *coproduct* of $\{A_\alpha\} \in \mathrm{ob}(\mathcal{C})$ consists of the data $A \in \mathrm{ob}(\mathcal{C})$ and morphisms $i_\alpha : A_\alpha \to A$ following the universal property— if $B \in \mathrm{ob}(\mathcal{C})$ with morphisms $f_\alpha : A_\alpha \to B$, there is a unique morphism $f : A \to B$ so that $f_\alpha = f \cdot i_\alpha$ for all α.

In this case, we write $A = \coprod_\alpha A_\alpha$. Note by the universal mapping property, if such exists, it is unique up to isomorphism.

The question of existence still exists. Here are some examples for where it holds.

Example 15.

1. In $\underline{\mathrm{Set}}$, coproducts exist and is just the disjoint union and the morphisms are simply the natural injections. This is why our notation of both the coproduct and disjoint union as $\coprod A_\alpha$ makes sense.

2. In $R-\underline{mod}$ we also have the existence of coproducts, but the construction is more complicated. Say $\{M_\alpha\}$ is an indexed set over I of R-modules. Let $M = \prod M_\alpha$ be the product. Set $\bigoplus M_\alpha = \{m : I \to \prod M_\alpha : m(\alpha) = 0 \text{ for all but finite } \alpha\}$. We again have the natural inclusion $i_\alpha : M_\alpha \to \bigoplus M_\alpha$. We claim this is the coproduct.

   Suppose $A \in \mathrm{ob}(R - \underline{mod})$ with morphisms $f_\alpha : M_\alpha \to A$. We define $f : \bigoplus M_\alpha \to A$ by defining $f(0, 0, \dots, m_\alpha, 0, 0, \dots)$ and then expanding linearlly because we are only dealing with finite sums. In fact, this is the only way for which $f \cdot i_\alpha = f_\alpha$. This shows $\bigoplus M_\alpha = \coprod M_\alpha$

   This example is particularly useful once we get to homological algebra. The similar construction also works for abelian groups (as they are $\mathbb{Z}$ modules) as well as arbitrary groups and rings.

1.6 Direct and Inverse Limits

> Why don't they call contravariant functors 'funky functors'?
>
> *Matt*

Let I be a directed set—a set I with a binary relation $\leq$ such that if $\alpha, \beta \in I$

1. $\alpha \leq \alpha$
2. $\alpha \leq \beta$ and $\beta \leq \gamma$ implies $\alpha \leq \gamma$.
3. Given $\alpha, \beta \in I$ there is $\gamma \in I$ such that $\alpha \leq \gamma$ and $\beta \leq \gamma$.

This defines a category $\underline{I}$ where $\mathrm{ob}(\underline{I}) = I$ and $\hom(A, B) = \begin{cases} \emptyset & \alpha \not\leq \beta \\ \{f_{\alpha,\beta}\} & \alpha \leq \beta \end{cases}$.

If $\mathcal{C}$ is any category, consider a convariant functor $F : \underline{I} \to \mathcal{C}$. Then F consists of a map $I \to \mathrm{ob}(\mathcal{C})$ and a map $(\alpha, \beta) \to \phi_{\alpha\beta} = F(f_{\alpha\beta})$ if $\alpha \leq \beta$.

So we have maps

$$A_\alpha \xrightarrow{\phi_{\alpha\beta}} A_\beta$$

when $\alpha \leq \beta$ such that $\phi_{\alpha\alpha} = \mathrm{id}_{F(\alpha)}$ and $\phi_{\beta\gamma} \cdot \phi_{\alpha\beta} = \phi_{\alpha\gamma}$ whenever $\alpha \leq \beta \leq \gamma$.

The "image" of F is called a *directed system* in $\mathcal{C}$ consisting of the data $\{A_\alpha, \phi_{\alpha\beta} : \alpha \leq \beta\}$. This allows us the view objects of $\mathcal{C}$ as living in a sort of lattic or structure mimicking the order of the directed set.

What happens then to the directed system as we move along the path of the system?

Definition 16. Let I be a directed set, $\mathcal{C}$ a category, $F : \underline{I} \to \mathcal{C}$ a covariant functor. Call $F(\alpha) = A_\alpha$ and $\phi_{\alpha\beta} = F(f_{\alpha\beta}) \in \hom(A_\alpha, A_\beta)$ for $\alpha \leq \beta$.

Then a *direct limit* $\lim_{\to_\alpha} A_\alpha = \lim_{\to_\alpha}(A_\alpha, \phi_{\alpha\beta})$ is a set $(A, \eta_\alpha)_{\alpha \in I}$ such that fo $A \in \mathrm{ob}(\mathcal{C})$ $\eta_\alpha \in \hom(A_\alpha, A)$ such that $\eta_\alpha = \eta_\beta \cdot \phi_{\alpha\beta}$.

$$\begin{tikzcd}
A_\alpha \arrow[rr, "\phi_{\alpha\beta}"] \arrow[dr, "\eta_\alpha"'] & & A_\beta \arrow[dl, "\eta_\beta"] \\
& A &
\end{tikzcd}$$

And we require A and $\{\eta_\alpha\}$ is universal in this construction—if $B \in \mathrm{ob}(\mathcal{C})$ and we have $\zeta_\alpha \in \hom(B, A_\alpha)$ satisfying the above commutative diagram, then there is a unique morphism $\theta : A \to B$ such that the following diagram commutes:

$$\begin{tikzcd}
A_\alpha \arrow[rr, "\phi_{\alpha\beta}"] \arrow[dr, "\eta_\alpha"] \arrow[ddr, "\zeta_\alpha"'] & & A_\beta \arrow[dl, "\eta_\beta"'] \arrow[ddl, "\zeta_\beta"] \\
& A \arrow[d, "\theta"] & \\
& B &
\end{tikzcd}$$

Therefore if directed limits exists, they are unique (up to isomorphism). Directed limits are also called *inductive limits*.

Here are some examples of when these direct limits exist:

Example 16. Let $\mathcal{C} = \underline{Set}$ and $I = \mathbb{N}$ with the usual ordering. The directed system take $i \in \mathbb{N}$ a set A_i so that the maps $A_i \hookrightarrow A_{i+k}$ is an injection. Thus we can view $A_i \subset A_{i+k}$

The directed limit is then the union $U_i A_i$ and the morphisms are just inclusions.

Example 17. In $R - \underline{mod}$, begin with a directed system of modules $\{M_\alpha, \phi_{\alpha\beta}\}$. Take $\widetilde{M} = \bigoplus_\alpha M_\alpha = \coprod_\alpha M_\alpha$. Let $N \subset \widetilde{M}$ be the submodule generated by sequences
$(0, 0, \cdots, m_\alpha, 0, \cdots, 0, -\phi_{\alpha\beta}(m_\alpha), 0, \cdots)$ where m_α appears in the α-th position and $-\phi_\alpha\beta(m_\alpha)$ in the β-th position.

Take $M = \widetilde{M}/N = \lim_{to_\alpha} M_\alpha$ where $\eta_\alpha \hookrightarrow \bigoplus_\alpha M_\alpha \to^p \widetilde{M}/N$.

This basically works by identifying objects that get mapped to each other.

Example 18. $\mathcal{C} = \underline{Grp}$ if we have a directed system $\{G_\alpha, \phi_{\alpha\beta}\}$ form $\coprod_\alpha G_\alpha$ in set. Put on an equivalence relation by $g_\alpha \in G_\alpha$ and $g_\beta \in G_\beta$ say $g_\alpha \sim g_\beta$ if there is $\alpha, \beta \leq \gamma$ so that $\phi_{\alpha\gamma}(g_\alpha) = \phi_{\beta\gamma}(g_\beta)$. Thus things are equivalent if, going far enough out, elements are mapped to the same thing.

Let $G = (\coprod G_\alpha)/\sim$ and $\eta_\alpha : G_\alpha \to G$ by $g_\alpha \mapsto [g_\alpha]$. Check the obvious group structure is well-defined and gives us a direct limit.

This same structure also works for $\underline{Ring}$ and such.

Definition 17. I directed set, $F : \underline{I} \to \mathcal{C}$ a contravariant functor. The "image" of F is called an *inverse system*. An *inverse limit* of an inverse system is an object $A \in \mathrm{ob}(\mathcal{C})$ along with morphisms $\eta_\alpha : A \to A_\alpha$ making the following diagram commute

$$\begin{tikzcd} A_\alpha \arrow[rr, "\phi_{\alpha\beta}"] & & A_\beta \\ & A \arrow[ul, "\eta_\alpha"] \arrow[ur, "\eta_\beta"'] & \end{tikzcd}$$

and satisfying the universal property that if B is another objects with morphisms $\zeta_\alpha : B \to A_\alpha$ making the above diagram commute then there is a morphism $\theta : B \to A$ so that $\zeta_\alpha = \eta_\alpha \circ \theta$.

We denote the inverse limit as $\lim_{\leftarrow_\alpha} A_\alpha$ or $\lim_{\leftarrow_\alpha} \{A_\alpha, \phi_{\beta\alpha}\}$.

Again, once we have existence, uniqueness follows from the universal mapping property. Inverse limits are also know as *projective limits*

An important example of this sort of limit is the *p-adic* integers.

Example 19. What do inverse limits look like in $\underline{\text{Grp}}$? It looks like Γ which consists of sequences in $\prod G_\alpha$ so that the elements of the sequence are compatible with the homomorphisms, i.e., $\phi_{\alpha\beta}(g_\alpha) = g_\beta$ for all $\alpha \geq \beta$. The maps $\eta_\alpha : G \to G_\alpha$ is projection on the α-th position.

Note then that the sequence of identities is in this group. Component-wise multiplication makes this into a group, as $\phi_{\alpha\beta}$ are homomorphisms.

Example 20. Fix G a group. Let $I = \{H \triangleleft G : (G : H) < \infty\}$. Order I by reverse inclusion (this works as $H \cap K \triangleleft G$ and is of finite index). Maps $\phi_{KH} : G/K \to G/H$ is then an inverse system in $\underline{\text{Grp}}$.

The inverse limit $\lim_{\leftarrow_H} G/H$ then exists, and the resulting group is called the *completion* of G. This is what is known as a *profinite group*—an inverse limit of finite groups.

We can embed $G \hookrightarrow \lim_{\leftarrow_H} G/H$ if $\bigcap_H H = \{e\}$.

1.7 The Hom Functors, Representable Functors, and Yoneda Lemma

> "That's why you should get a Mastercard"
>
> James in response to "One student is still having visa problems"

Given a category $\mathcal{C}$, there are two types of natural functors $\mathcal{C} \to \underline{Set}$. The first is covariant: $\hom(A, -)$ for fixed $A \in \text{ob}(\mathcal{C})$. The second is contravariant $\hom(-, A)$.

N.B. We can also view this is a bi-functor $\hom_{\mathcal{C}}(-,-) : \mathcal{C}^{op} \times \mathcal{C} \to \underline{Set}$.

Definition 18. A covariant functor $F : \mathcal{C} \to \underline{Set}$ is called *representable* if it is representable as a hom functor, i.e., $F \cong \hom(A, -)$ for some $A \in \text{ob}(\mathcal{C})$.

Similarly, a contravariant functor $G : \mathcal{C} \to \underline{Set}$ is *representable* if naturally isomorphic to $\hom(-, A)$ for some $A \in \text{ob}(\mathcal{C})$.

In either case, we call A a *representative* of the functor.

This definition leads to the following question: In what cases do we have a natural transformation $\eta : \hom(A, -) \to F$ for a covariant functor $F : \mathcal{C} \to \underline{Set}$?

If such η exsits, we always have a set map $\eta_B : \hom(A, B) \to F(B)$ such that for all $g \in \hom(B, C)$ we have the commuting diagram

$$\begin{tikzcd} \hom(A,B) \arrow[r, "\eta_B"] \arrow[d, "\hom(A,g)"] & F(B) \arrow[d, "F(g)"] \\ \hom(A,C) \arrow[r, "\eta_C"] & F(C) \end{tikzcd}$$

We also have an evaluation map for any $a \in A$ and $B \in \mathrm{ob}(\mathcal{C})$ by $a_B : \hom(A,B) \to F(B)$ by $f \mapsto F(f)(a)$. Thus for this natural transformation to exists, we must have commutivity of the diagram satisfied by the evaluation maps. If $\eta_B = a_B$ then commutativity means for all $f \in \hom(A,B)$ and $g \in \hom(B,C)$, $F(gf)(a) = F(g)(F(f)(a)) = F(g)(a_B(f)) = a_C \hom(A,g)(f) = F(gf)(a)$. This shows $\eta(a) : B \to a_B$ is a natural transformation from $\hom(A,-) \to F$.

N.B. $\eta(a)_A(\mathrm{id}_A) = a$.

It turns out these are all the natural transformations we can get from $\hom(A,-)$ to F.

Lemma 1. *If $\eta : \hom(A,-) \to F$ is a natural transformation, then $\eta = \eta(a)$ for some $a \in F(A)$.*

Proof. Consider $f \in \hom(A,B)$ and the diagram

$$\begin{tikzcd} \hom(A,A) \arrow[r, "\eta_A"] \arrow[d, "\hom(A,f)"] & F(A) \arrow[d, "F(f)"] \\ \hom(A,B) \arrow[r, "\eta_B"] & F(B) \end{tikzcd}$$

Then

$$\eta_B(f) = \eta_B(f \cdot \mathrm{id}_A) = \eta_B(\hom(A,f)(\mathrm{id}_A) = F(f) \cdot \eta_A(\mathrm{id}_A)$$

Take $a = \eta_A(\mathrm{id}_A)$. □

Combining this lemma with the previous discussion, we arrive at Yoneda's Lemma:

Lemma 2 (Yoneda's Lemma)**.** *Let $F : \mathcal{C} \to \underline{Set}$ be a covariant functor and $A \in \mathrm{ob}(\mathcal{C})$. Then the natural transformations of $\hom(A,-)$ to F are in bijection with elements in $F(A)$ by $a \in F(A) \mapsto \eta$ where $\eta_B = a_B$. The inverse is given my $\eta \mapsto \eta_A(\mathrm{id}_A)$.*

If F is representable, with representative A, the natural isomorphism is given by $\eta(a)$ for some $a \in F(A)$. So we can call (A,a) a *representative* of F.

1.8 The Category of Modules

> "I've done almost none of the analysis." "That's a good move."
>
> Cogdell in response to Daniel

Let R be a ring (with identity). We have seen $R - \underline{mod} \cong \underline{mod} - R^{op}$. For this reason we will focus only on left R-modules. This category has a 0-object which is the trivial module. For $M, N \in \mathrm{ob}(\mathcal{C})$ we define the 0-*morphism* to be the composition of $M \to 0 \to N$. In this context, the map sends everything in M to 0_N.

Importantly, each of the morphism sets $\hom_R(M,N)$ carry the addition structure of an abelian group. For $f, g \in \hom_R(M,N)$ we define $f + g \in \hom_R(M,N)$ by $(f+g)(m) = f(m) + g(m)$. The identity in the group is given by the 0-morphism. Thus $(\hom_R(M,N), +, 0)$ is an abelian

group. Moreover, the addition commutes with composition (i.e. $f \cdot (g+h) = f \cdot g + f \cdot h$ and $(f+g) \cdot h = f \cdot h + g \cdot h$ whenever these compostions make sense).

When looking at functors between module categories, we would also like to consider when the functor preserves the group structures on the morphism sets.

Definition 19. A functor $F : R - \underline{mod} \to S - \underline{mod}$ is *additive* if $F(f+g) = F(f) + F(g)$ for $f, g \in \hom_R(M, N)$.

This can be said of both covariant and contravariant functors.

From this point forward, all functors between module categories are assumed to be additive.

N.B. In the case $M = N = P$, then $End(M) := \hom_R(M, M)$ gives a ring structure where multiplication is defined by composition.

Recall that in $R-\underline{mod}$ we have both product and coproducts (because these are abelian groups). In the case the index set I of the (co)product is finite, the product and coproduct over I is the same group, and the injections $i_j : M_j \to M$ and projections $p_i : M \to M_i$. Then $p_j \circ i_j = \mathrm{id}_{M_j}$, $p_k \circ i_j : M_j \to M_k = 0$, and $\sum_{i=1}^n i_i p_i = \mathrm{id}_M$.

Moreover, these properties of morphisms completely determine M. As if we have a module M' with injections $i'_j \in \hom(M_i, M')$ and projections $p'_j \in \hom(M', M_j)$ satisfying the above three properties, then $\theta = \sum_{j=1}^n i'_j p_j \in \hom(M, M')$ and $\theta' = \sum_{j=1}^n i_j p'_j \in \hom(M', M)$. Then $\theta' \cdot \theta \in \hom(M, M)$ is the identity on M and $\theta \cdot \theta'$ is the identity on M'. Therefore the morphisms $\{i_j\}, \{p_j\}$ completely characterize the product and coproduct.

A consequence of this fact is that an additive functor preserves products and coproducts. We can express this as $F(\oplus_{i=1}^n M_i) = \oplus_{i=1}^n F(M_i)$ with projections $F(p_j)$ and injections $F(i_j)$ (in the case of a contravariant functor, the projections are sent to the injections and vice versa).

From these properties of modules, we can define an *additive category*

Definition 20. A category $\mathcal{C}$ is called an *additive category* if

1. $\mathcal{C}$ has a 0-object.
2. For $M, N \in \mathrm{ob}(\mathcal{C})$ we have $\hom_{\mathcal{C}}(M, N)$ has the structure of abelian groups with the identity being the zero morphism.
3. The additive structure distributes over composition ($f \cdot (g+h) = f \cdot g + f \cdot h$ and $(g+h) \cdot f = g \cdot f + h \cdot f$ whenever these compositions are defined.)
4. Given a finite set of objects $A_1, \ldots, A_n \in \mathrm{ob}(\mathcal{C})$ there exists $A \in \mathrm{ob}(\mathcal{C})$ and morphisms $p_j \in \hom_{\mathcal{C}}(A, A_j)$ and $i_j \in \hom_{\mathcal{C}}(A_j, A)$ such that $p_j \cdot i_j = \mathrm{id}_{A_j}$, $p_j \cdot i_k = 0$ for $j \neq k$ and $\sum_{j=1}^N i_j \cdot p_j = \mathrm{id}_A$

1.8.1 Exactness

If M, N, P are R-modules and we have maps

$$M \xrightarrow{f} N \xrightarrow{g} P$$

Is *exact* at N if $\mathrm{im}(f) = \ker(g)$.

The definition extends to longer sequences

$$\cdot \longrightarrow M_1 \xrightarrow{f_1} M_2 \xrightarrow{f_2} M_3 \xrightarrow{f_3} M_4 \xrightarrow{f_4} \cdots$$

is *exact* at M_i if $\mathrm{im}(f_{i-1}) = \ker(f_i)$.

Proposition 3. *1. $0 \to M \xrightarrow{f} N$ is exact at M if and only if $\ker(f) = 0$ if and only if f is injective if and only if f is monic.*

2. $M \xrightarrow{g} N \to 0$ is exact at N iff $\mathrm{im}(g) = N$ iff f is surjective iff f is epic.

3. The map $0 \to \ker(f) \to M \xrightarrow{f} N$ is always exact.

4. If we have $M \xrightarrow{f} N$, define $\mathrm{coker}(f) := N/\mathrm{im}(f)$. Then $M \xrightarrow{f} N \to \mathrm{coker}(f) \to 0$ is always exact. This gives f surjective iff $\mathrm{coker}(f) = 0$.

5. If $f : M \to N$ is any morphism in $R - \underline{mod}$, we can always factor $M \to \mathrm{im}(f) \to N$ where the first map is epic and the second is monic.

6. $0 \to \ker(f) \to M \xrightarrow{f} N \to \mathrm{coker}(f) \to 0$ is exact.

Definition 21. An additive category $\mathcal{C}$ is an *abelian* category if

1. Every morphism in $\mathcal{C}$ has a kernel and cokernel.
2. Every monic morphism is a kernel of its cokernel.
3. Every epic morphism is a cokernel of its kernel.
4. Every morphism factors as the composition of an epic followed by a monic.

2 Module Categories

2.1 Exactness and Functors

Let $F : R - \underline{mod} \to S - \underline{mod}$ be an additive functor. We say F is *exact* if is preserves short exact sequences. This means if

$$0 \to M' \to M \to M'' \to 0$$

is exact then so is

$$0 \to F(M') \to F(M) \to F(M'') \to 0$$

(in the case of a contravariant functor the last row of arrows are reversed).

Often, it is enough to consider when a funtor is partially exact, i.e., exact on one side.

If F is covariant, then F is *right exact* if

$$M' \to M \to M'' \to 0$$

exact implies

$$F(M') \to F(M) \to F(M'') \to 0$$

is exact. Similarly, F is *left exact* if

$$0 \to M' \to M \to M''$$

exact implies

$$0 \to F(M') \to F(M) \to F(M'')$$

is exact.

In the case F is contravariant, F is *right exact* if

$$0 \to M' \to M \to M''$$

implies

$$F(M'') \to F(M) \to F(M') \to 0$$

is exact. Similarly, F is called *left exact* if

$$M' \to M \to M'' \to 0$$

implies

$$0 \to F(M'') \to F(M) \to F(M')$$

is exact.

For module cateogries, we have two natural functors

$$\hom(M, -) : R - \underline{mod} \to \mathbb{Z} - \underline{mod}$$

and

$$\hom(-, M) : R - \underline{mod} \to \mathbb{Z} - \underline{mod}$$

Theorem 1. *These functors are left exact.*

Proof. We will prove this for $\hom(M, -)$, and the proof for the contravariant Hom functor is left for homework.

Suppose

$$0 \to N' \xrightarrow{f} N \xrightarrow{g} N''$$

is an exact sequence in $R - \underline{mod}$. Thus f is injective and $\mathrm{im}(f) = \ker(g)$. We first show $\hom(M, f)$ is injective. Indeed, if $h, h' \in \hom(M, N')$ with $f \circ h = f \circ h'$, we have $h = h'$, as f is injective.

Now suppose $h \in \hom(M, N) \in \ker(\hom(M, g))$. Thus, $\mathrm{im}(h) \subseteq \ker(g)$ which implies $\mathrm{im}(h) \subset \mathrm{im}(f)$. As f is injective, for every $m \in M$, we can find a unique $n' \in N'$ so that $h(m) = f(n')$. Thus define a function $h' : M \to N'$ by $m \mapsto n'$. It is easy to check this is a module homomorphism and that $f \circ h' = h$. Thus $h \in \mathrm{im}(\hom(M, f))$.

Now suppose $h \in \mathrm{im}(\hom(M, f))$. thus $h = f \circ h'$ for some morphism $h' : M \to N'$. Then $g \circ h = g \circ f \circ h' = 0 \circ h' = 0$. Thus $\mathrm{im}(\hom(M, f)) \subseteq \ker(\hom(M, g))$. The reverse containments give $\ker(\hom(M, g)) = \mathrm{im}(\hom(M, f))$.

This shows the Hom functor is left exact. □

In the homework, we will also see examples showing these Hom functors are strictly left exact and not exact.

N.B. In the case R is commutative, the $\hom_R(M, N)$ sets also have the structure of an R-module given by $(r \cdot f)(m) = r \cdot f(m)$. Thus we can view $\hom_R(M, -)$ and $\hom_R(-, M)$ as functors to $R - \underline{mod}$ and not just $\mathbb{Z} - \underline{mod}$.

The next example of module functors we consider are the tensor functors $\otimes$—we will see these are right exact.

Definition 22. Let M be a right R-module and N be a left R-module. Define a *Balanced Product* of M, N to be an abelian group P with a map $f : M \times N \to P$ such that

1. $f(m_1 + m_2, n) = f(m_1, n) + f(m_2, n)$
2. $f(m, n_1 + n_2) = f(m, n_1) + f(m, n_2)$
3. $f(mr, n) = f(m, rn)$

A morphism between balanced products of M and N consists of a morphism of abelian groups $\eta : (P, f) \to (Q, g)$ such that $\eta(f(m, n)) = g(m, n)$. Thus the following diagram commutes

$$\begin{tikzcd} M \times N \arrow[r, "f"] \arrow[d, "g"] & P \arrow[dl, "\eta"] \\ Q & \end{tikzcd}$$

A *tensor product* $(M \otimes_R N, \otimes)$ is a balanced product with the following universal mapping property: If (P, f) is any other balanced product of M, N then there is a unique map $\eta : M \otimes_R N \to P$ such that the following diagram commutes

$$\begin{tikzcd} M \times N \arrow[r, "\otimes"] \arrow[d, "f"] & M \otimes_R N \arrow[dl, "\eta"] \\ P & \end{tikzcd}$$

We note that $M \otimes N$ exists, and the universal mapping property gives it is unique up to isomorphism. Also, $M \otimes N = \langle m \otimes n : m \in M \;\; n \in N \rangle$. Elements of the form $m \otimes n$ are caled *pure tensors* or *decomposable elements*.

2.2 Tensor Products

> "I believe you beat that horse with Gaith"
>
> Cogdell

Recall the construction of the tensor product: First, form the free $\mathbb{Z}$-module on $M \times N$. This is just $F = \bigoplus_{M \times N} \mathbb{Z}$. Elements of F take the form $\sum x_i(m_i, n_i)$ where $m_i \in M$, $n_i \in N$ and $x_i \in \mathbb{Z}$. Let A be the submodule of F generated by the relations defining the balanced product: $(m_1 + m_2, n) - (m_1, n) - (m_2, n)$, $(m, n_1 + n_2) - (m, n_1) - (m, n_2)$, and $(mr, n) - (m, rn)$. Then $M \otimes_R N = F/A$ and $m \otimes n = (m, n) + A$.

2.2.1 $\otimes$ as a functor

We have $\otimes : \underline{mod} - R \times R - \underline{mod} \to \mathbb{Z} - \underline{mod}$. If $M \xrightarrow{f} M'$ and $N \xrightarrow{g} N'$ then we have a map

$$M \times N \xrightarrow{f \times g} M' \times N' \xrightarrow{\otimes} M' \otimes N'$$

The composition gives a balanced product on $M \times N$, so it induces a morphism

$$M \otimes N \xrightarrow{f \otimes g} M' \otimes N'$$

On pure tensors, $(f \otimes g)(m \otimes n) = f(m) \otimes g(n)$. In the case we have multiple morphisms

$$M \xrightarrow{f} M' \xrightarrow{f'} M''$$

$$N \xrightarrow{g} N' \xrightarrow{g'} N''$$

Then on the generators, we have

$$(f' \cdot f)(\otimes(g' \cdot g)(m \otimes n) = (f'f)(m) \otimes (g'g)(n) = (f' \otimes g') \cdot (f \otimes g)(m\otimes)$$

Whence

$$(f' \cdot f) \otimes (g' \cdot g) = (f' \otimes g') \cdot (f \otimes g)$$

Theorem 2. *The maps $(M \times N) \to M \otimes N$ and $(f \times g) \mapsto f \otimes g$ defines a covariant bifunctor $\underline{mod} - R \times R - \underline{mod} \to \mathbb{Z} - \underline{mod}$.*

By fixing either M or N we get a tensor functor $M \otimes - : R - \underline{mod} \to \mathbb{Z} - \underline{mod}$ or a tensor functor $- \otimes N : \underline{mod} - R \to \mathbb{Z} - \underline{mod}$ respectively.

We should also check these functors are additive: Indeed, on generators we have $(f_1 + f_2) \otimes \mathrm{id}_N(m \otimes n) = (f_1 + f_2)(m) \otimes n = f_1(m) \otimes n + f_2(m) \otimes n$ which shows $(M \otimes -)(f_1 + f_2) = (M \otimes -)(f_1) + (M \otimes -)(f_2)$. A similar proof demonstrates the same for $(- \otimes N)$.

This means the $\otimes$ functors preserve finite products and coproducts. However, there is an issue for arbitrary products. This problem does not arise for arbitrary coproducts.

Proposition 4. *If $\{M_\alpha\}_{\alpha \in I}$ a set of right R-modules, and N a fixed left R-modules, then $(\oplus_{\alpha \in I} M_\alpha) \otimes N \cong \oplus_\alpha (M_\alpha \otimes N)$ by $((x_\alpha)_{\alpha \in I} \otimes n) \mapsto (x_\alpha \otimes n)_{\alpha \in I}$.*

Proof. We claim that the defined map is a balanced product, so we have an induced map

$$(\oplus M_\alpha) \otimes N \xrightarrow{\eta_N} \oplus_\alpha (M_\alpha \otimes N)$$

In the other direction, we have the injections $i_\alpha : M_\alpha \to \oplus_\alpha M_\alpha$. Tensoring with the identity gives

$$i_\alpha \otimes \mathrm{id}_N : M_\alpha \otimes N \to (\oplus_\alpha M_\alpha) \otimes M$$

Using the universal mapping property of the coproduct gives a map

$$\oplus_\alpha (M_\alpha \otimes N) \to (\oplus_\alpha M_\alpha) \otimes N$$

One easily checks this is the inverse of η_N (by checking on generators). $\square$

2.2.2 Exactness of $\otimes$ functors

Theorem 3. *The one-sided $\otimes$ functors are both right exact, covariant functors.*

Proof. Consider $M \otimes - : R - \underline{mod} \to \mathbb{Z} - \underline{mod}$. Suppose

$$N' \xrightarrow{f} N \xrightarrow{g} N'' \to 0$$

is an exact sequence.

Consider the exactness of

$$M \otimes N' \xrightarrow{\mathrm{id}_M \otimes f} M \otimes N \xrightarrow{\mathrm{id}_M \otimes g} M \otimes N'' \to 0$$

Suppose $m \otimes n''$ is a pure tensor of $M \otimes N''$. As g is surjective, there is $n \in N$ such that $g(n) = n''$. Thus $\mathrm{id}_M \otimes g(m \otimes n) = m \otimes n''$. This shows $\mathrm{id}_M \otimes g$ is surjective.

Now suppose $\sum m_i \otimes f(n_i')$ is in the image of $\mathrm{id}_M \otimes f$. We can calculate that $(\mathrm{id}_M \otimes g)(\sum m_i \otimes f(n_i')) = \sum m_i \otimes gf(n_i') = \sum m_i \otimes 0 = 0$. Therefore $\mathrm{im}(\mathrm{id}_M \ \ otimesf) \subset \ker(\mathrm{id}_M \otimes g)$.

As $\mathrm{id}_M \otimes g$ is surjective, $M \otimes N'' \cong M \otimes N / \ker(\mathrm{id}_M \otimes g)$ and as $\mathrm{im}(\mathrm{id}_M \otimes f) \subseteq \ker(\mathrm{id}_M \otimes g)$ we can map

$$M \otimes N / \mathrm{im}(\mathrm{id}_M \otimes f) \xrightarrow{\theta} M \otimes N'' \to 0$$

In this case, the statement $\mathrm{im}(\mathrm{id}_M \otimes f) = \ker(\mathrm{id}_M \otimes g)$ is the same as θ being an isomorphism. Therefore, we wil show θ is an isomorphism, by constructing θ' its inverse.

For $M \times N''$, we can define $\theta'(m, n'') = m \otimes n + \mathrm{im}(\mathrm{id}_M \otimes f)$ where $n \in N$ is some element for which $g(n) = n''$. This is well-defined as if $n' \in N$ is another such element, $m \otimes n - m \otimes n' \in \mathrm{im}(\mathrm{id}_M \otimes f)$ by the exactness of

$$N' \xrightarrow{f} N \xrightarrow{g} N'' \to 0$$

This is clearly a bilinear map, and so we have an induced map

$$\theta : M \otimes N'' \to M \otimes N / \mathrm{im}(\mathrm{id}_M \otimes f)$$

such that on pure tensors, the map is analogous to the one described above.

To see this is the inverse of θ, for $m \otimes n + \mathrm{im}(\mathrm{id}_M \otimes f)$

$$\theta'(\theta(m \otimes n + \mathrm{im}(\mathrm{id}_M \otimes f))) = \theta'(m \otimes g(n) = m \otimes n$$

and

$$\theta(\theta'(m \otimes n'')) = \theta(m \otimes n + \ker(\mathrm{id}_M \otimes f)) = m \otimes g(n) = m \otimes n''$$

Therefore, these maps are inverses on the generators, so inverses on the whole modules.

This shows that $M \otimes -$ is right exact. □

In general, these tensors functors are not exact. For this we would require $M \otimes -$ to preserve injectivity. When this is the case, we call M *flat*.

Definition 23. A right R-module M is called *flat* if

$$0 \to N' \xrightarrow{f} N$$

exact implies

$$0 \to M \otimes N' \xrightarrow{\mathrm{id}_M \otimes f} M \otimes N$$

is exact.

Proposition 5. *$M \otimes -$ is exact if and only if M is flat.*

Similarly, $- \otimes M$ is exact if and only if M is flat (where the definition of flat is generalized for left R-modules).

N.B. In the case R is commutative, if M is right R-module then we can put the structure of a left R-module on M by $r \cdot m = m \cdot r$. The same gives a right R-module action from a left.

In this case we can compute $M \otimes N$ for any type of R-modules M and N. Also, $M \otimes N$ has the structure of an R-module by

$$r \cdot (m \otimes n) = (r \cdot m) \otimes n = m \otimes (r \cdot n)$$

This makes $\otimes$ a bifunctor from $R - \underline{mod} \times R - \underline{mod} \to R - \underline{mod}$. This is a canonical example of a $\otimes$-*category*—a category $\mathcal{C}$ with a bi-functor $\otimes : \mathcal{C} \otimes \mathcal{C} \to \mathcal{C}$ satisfying some axioms.

2.3 Projective and Injective Modules

> "Did you think anymore about the product of two additive categories being additive?"
> "No."
> "Neither did I."
>
> Cogdell and Vilas

As we will see projective modules are generalization of free modules, we will first look at properties of free mdoules over a ring R.

Definition 24 (Free modules by Universal Mapping Property)**.** Let X be a set. The *free R-module generated by X* is an R-module F_X with an embedding

$$u : X \hookrightarrow F_X$$

such that for any set map $\phi : X \to M$ (M and R-module) there is a unique R-linear map $\widetilde{\phi} : F_X \to M$ such that $\phi = \widetilde{\phi} \cdot u$.

The construction of such an object is given by $F_X = \bigoplus_{x \in X} R$. We have $u : X \to F_X$ by $u(x) = (0, 0, \dots, 1, 0, \dots)$ where 1 occurs in the place corresponding to x. Given $\phi : X \to M$ the induced map is given by $\widetilde{\phi}(u(x)) = \phi(x)$ and then extended uniquely by linearity.

From the construction we see F_X is determined by the cardinality of X.

2.4 Free Modules

There are 4 basic properties of free modules:

1. Every module is a quotient of a free module.

   Proof. Take for X a generating set for M. Take $F = F_X$ with the embedding $\phi : X \hookrightarrow F_X$. By the universal property, there is a module morphism $\widetilde{\phi} : F_X \to M$. As M is generated by X, this morphism is surjective. □

2. Any morphism to a quotient can be lifted. This means

$$\begin{tikzcd} & F \arrow[d, "f"] \arrow[dl, dotted, "h"'] & \\ M \arrow[r, "g"'] & M' \arrow[r] & 0 \end{tikzcd}$$

   Proof. Let F be a free module in X. For $\bar{x} \in X$, $f(\bar{x}) = m''_X \in M''$. As g is surjective, we pull back to m_X with $g(m_X) = m''_X$. Define $\phi(\bar{x}) = m_X$. Then by the universal mapping property of F_X there is a module morphism $h = \widetilde{\phi}$, and this makes the diagram commute. □

3. The convariant Hom functor $\hom_R(F, -)$ is exact.

Proof. Begin with

$$0 \to M' \to M \xrightarrow{g} M'' \to 0$$

exact. As $\hom(F, -)$ is left exact, we need only check

$$\hom(F, M) \xrightarrow{\hom(F,g)} \hom(F, M'') \to 0$$

is surjective. But this exactly the fact that morphisms onto quotients can be lifted. □

4. Any short exact sequence ending in F splits.

   Proof. Let

   $$0 \to M' \to M \xrightarrow{g} F \to 0$$

   be exact. We want a morphism $s : F \to M$ so that $g \cdot s = \mathrm{id}_F$. Again this comes from 2—take $\mathrm{id} : F \to F$ and as $M \xrightarrow{g} F \to 0$ gives a quotient, we can lift g to a morphism $s : F \to M$. □

2.4.1 Projective Modules

We can generalize some the properties of free modules to general categories to get projective modules. The most important property of the previous 4 basic facts was that of lifting a map onto a quotient. As lifting can be defined just in terms of morphisms, this can be generalized to arbitrary categories.

Definition 25. An R-module P is *projective* if we have the following commutative diagram in R-<u>mod</u>

$$\begin{tikzcd} & P \arrow[dl, dotted, "h"'] \arrow[d, "f"] & \\ M \arrow[r, "g"] & M'' \arrow[r] & 0 \end{tikzcd}$$

There are 4 equivalent properties of projective modules.

Proposition 6. *The following are equivalent.*

1. *We have the following commutative diagram*

$$\begin{tikzcd} & P \arrow[dl, dotted, "h"'] \arrow[d, "f"] & \\ M \arrow[r, "g"] & M'' \arrow[r] & 0 \end{tikzcd}$$

2. *Every short exact sequence ending in P splits.*

3. *P is a direct summand of a free module. This means there exists N such that $N \oplus P = F$ is free.*

4. *The covariant Hom functor $\hom_R(P, -)$ is exact.*

Proof. $1 \Rightarrow 2$ Lift the identity morphism $\mathrm{id}_P : P \to P$ through $g : M \to P$.

$2 \Rightarrow 3$ As P is the quotient of a free module we have

$$0 \to N \to F \to P \to 0$$

exact. This sequence splits, so $F \cong N \oplus P$.

$3 \Rightarrow 1$ Write $F = N \oplus P$. Suppose we have a diagram

$$\begin{tikzcd} & P \arrow[d, "f"] & \\ M \arrow[r, "g"] & M'' \arrow[r] & 0 \end{tikzcd}$$

with the bottom row exact.

Then we have

$$\begin{tikzcd} 0 \arrow[r] & N \arrow[r] & F \arrow[r, "p"] \arrow[d, dotted, "h'"] & P \arrow[l, bend right, "s"'] \arrow[r] \arrow[d, "f"] & 0 \\ & & M \arrow[r, "g"] & M'' \arrow[r] & 0 \end{tikzcd}$$

From lifting $f \cdot p$ through g.

But we have a section $s : P \to F$, and the map $s \cdot h'$ is what we want.

$1 \Rightarrow 4$ This is exactly the same as in free modules: given

$$0 \to M' \to M \to M'' \to 0$$

we need only check

$$\hom(P, M) \to \hom(P, M'') \to 0$$

is surjective, which is the case from lifting.

$4 \Rightarrow 1$ This follows from $\hom(P, g)$ is surjective onto $\hom(P, M'')$—the lift is the preimage of $f : P \to M''$ under this map $\square$

From the fact that free modules are projective, and every module is the quotient of a free module, every module M is the quotient of a projective module. We usually phrase this as "R-<u>mod</u> has enough projectives"

This is important in homological algebra when we want to express things in terms of projective resolutions.

Proposition 7. *Every projective module P is flat.*

N.B. Note this uses exercise 19 in the homework: If I is an index set, then $N = \oplus_{\alpha \in I}$ is flat if and only if each N_α is flat for left R-modules.

Proof. We first cliam that R is flat as an R-module. Indeed, $N \otimes R \cong N$ so the image of an exact sequence is the same sequence.

Next, we claim that any free R-module is flat. This follows from the above note.

Finally, as P is the direct summand of a free module, the note above gives P is flat. $\square$

2.4.2 Aside: Exactness in abelian categories

$$0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$$

In the case of modules, exactness at B means $\mathrm{im}(f) = \ker(g)$. For general categories, we then say exactness at B is $g \circ f = 0$ (because we have the notion of the zero map in abelian categories with the zero object) and if $k : K \to B$ is the kernel of g and $c : B \to F$ the cokernel of f, $c \circ k = 0$. The first condition is equivalent to $\mathrm{im}(f) \subset \ker(g)$, while the second is equivalent to $\ker(g) \subset \mathrm{im}(f)$.

2.5 Injective Modules

> "So that's what Folland looks like these days"
> "That's a book, not Folland."
>
> Cogdell and Nicolas Bolle

Injective modules historically preceeded projectives and are the categorical duals of projective modules and relate to the possibility of *dividing* by elements of R.

Definition 26. An R-module Q is *injective* if whenver we have a diagram that is exact on the top row

$$\begin{tikzcd} 0 \arrow[r] & M' \arrow[r, "g"] \arrow[d, "f"] & M \\ & Q & \end{tikzcd}$$

We can extend f to a morphism

$$\begin{tikzcd} 0 \arrow[r] & M' \arrow[r, "g"] \arrow[d, "f"] & M \arrow[dl, dotted, "h"] \\ & Q & \end{tikzcd}$$

making the diagram commutes.

One can think of this in the following way: M' sits inside of M as a submodule by the exactness of the top diagram, so Q injective means we can extend the map f to all of M.

Proposition 8. *The following are equivalent:*

1. *Q is injective*
2. *The contravariant Hom functor* $\hom(-, Q)$ *is exact*
3. *Any exact sequence*
$$0 \to Q \to M \to M' \to 0$$
splits

Proof. $1 \Leftrightarrow 2$: One can just dualize the statement for projectives, or: We have $\hom(-, Q)$ is left exact, thus we must show that given

$$0 \to M' \xrightarrow{f} M$$

exact we have $\hom(M, Q) \to \hom(M', Q)$ surjective. This follows immediately from the ability to lift morphisms given in the definition.

Likewise, the fact that we can lift is exactly that $\hom(M, Q) \to \hom(M', Q)$ is surjective.

$1 \Rightarrow 3$: Given

$$0 \to Q \to M \to M' \to 0$$

lift the morphism $\mathrm{id}_Q : Q \to Q$.

$3 \Rightarrow 1$: We first need the following:

Theorem 4. *Every module is the submodule of an injective.*

This is the analogue of projectives being a direct summand of a free module.

Proof. We utilize the connection between injectivity and divisibility. We first do the case of $\mathbb{Z}$-modules, in which case the notion of divisibility is a divisible abelian group (every map $t \mapsto mt$ for $m \in \mathbb{Z}$ non-zero is surjective).

Proposition 9. *If T is a divisible group then it is injective.*

Proof. Consider a diagram of abelian groups

$$\begin{tikzcd} 0 \arrow[r] & M' \arrow[r, "i"] \arrow[d, "f"] & M \arrow[dl, dotted, "h"] \\ & T & \end{tikzcd}$$

where we want to construct the dotted morphism. For $x \in M$, $x \notin M'$, if x is free over M' ($\langle x \rangle \cap M' = \emptyset$) we can extend f to x by choosing $t \in T$ and defining $h(x) = t$. In the case x is not free, take n to be the smallest positive integer with $nx \in M'$. Then $f(nx)$ exists and is in T. As T is divisible, we can choose $u \in T$ with $nu = f(nx)$. We then define $h(x) = u$.

In this way, we can always expand h from f one element at a time, and it is easy to show this extension is well-defined.

Use Zorn's Lemma to construct a morphism g that extends f to a maximal subgroup of M (The poset will be a pair $(f_{M''}, M'')$ with $(f_{M''}, M'') \leq (f_H, H)$ if both $M'' \subseteq H$ and $f_{M''} = f_{H|M''}$—the maximal element of the chain is just the function on the union of all the sets, and the funtion taking an element to where any function in the chain (defined on the element) sends the element). Because we can extend any morphism on a proper subset of M, the maximally defined function must be defined on all of M.

This shows T is injective. □

We finish our claim for abelian groups.

Proposition 10. *Any abelian group is a subgroup of a divisible group.*

Proof. We must first define a notion of dual for an abelian group. Define $M^* = \hom_{\mathbb{Z}}(M, \mathbb{Q}/\mathbb{Z})$. If F is a free abelian group, than the dual is divisible. Indeed, any element on the dual is defined by the action of the function on its generators, and we can divide these images by any integer in $\mathbb{Q}/\mathbb{Z}$.

Now let M be an abelian group, and M^* its dual. Thus M^* is a quotient of a free group

$$F \to M^* \to 0$$

applying duality again yields

$$0 \to M^{**} \to F^*$$

an exact sequence. We claim $M \hookrightarrow M^{**}$ by the evaluation map. This is equivalent to saying that for every $m \in M$ with $m \neq 0$ there is a linear funtion $M \to \mathbb{Q}/\mathbb{Z}$ on which the image of m is non-zero. Again we utilize the divisibilty of $\mathbb{Q}/\mathbb{Z}$ to extend a map $\langle m \rangle \to \mathbb{Q}\mathbb{Z}$ to all of M. Thus we are left with finding a non-zero morphism $\langle m \rangle \to \mathbb{Q}/\mathbb{Z}$—iff m has torsion, send it to $\frac{1}{k} + \mathbb{Z}$ where k is the torsion, and if not, send it to $\frac{1}{2} + \mathbb{Z}$. Thus we have $M \hookrightarrow F^{**}$ where the latter group is divisible and thus injective. $\square$

Now we extend this to R-modules. There are 4 steps:

1. If R is any ring and A an abelian group, then $\hom_{\mathbb{Z}}(R, A)$ is naturally a left R module where the module structure is given by $r \cdot f(x) = f(xr)$[4].

2. If M is any R-module, there exists a natural isomorphism of abelian groups

$$\hom_{\mathbb{Z}}(M, A) \cong \hom_R(M, \hom_{\mathbb{Z}}(R, A))$$

given by $psi \mapsto f_\psi$ where $f_\psi(m)(r) = \psi(rm)$. The inverse is given by $f \mapsto \psi_f$ where $\psi_f(m) = f(m)(1)$. We interpret this as a natural isomorphism between $\hom_{\mathbb{Z}}(-, A)$ and $\hom_R(-, \hom_{\mathbb{Z}}(R, A))$.

3. If T is a divisible abelian group, then $\hom_{\mathbb{Z}}(R, T)$ is injective as an R-module. We show that $\hom_R(-, \hom_{\mathbb{Z}}(R, T))$ is an exact functor. Take an exact sequence

$$0 \to M' \to M \to M'' \to 0$$

and get

$$0 \to \hom_R(M'', \hom_{\mathbb{Z}}(R, T)) \to \hom_R(M, \hom_{\mathbb{Z}}(R, T)) \to \hom_R(M', \hom_{\mathbb{Z}}(R, T)) \to$$

with the left portion exact. By the natural isomorphism from the previous step

$$\begin{tikzcd}
0 \arrow[r] & \hom_R(M'', \hom_{\mathbb{Z}}(R, T)) \arrow[r] \arrow[d, leftrightarrow] & \hom_R(M, \hom_{\mathbb{Z}}(R, T)) \arrow[r] \arrow[d, leftrightarrow] & \hom_R(M', \hom_{\mathbb{Z}}(R, T)) \arrow[r] \arrow[d, leftrightarrow] & 0 \\
0 \arrow[r] & \hom_{\mathbb{Z}}(M'', T) \arrow[r] & \hom_{\mathbb{Z}}(M, T) \arrow[r] & \hom_{\mathbb{Z}}(M'', T) \arrow[r] & 0
\end{tikzcd}$$

The bottom sequence is exact because T is injective and hence we have surjectivity at $\hom_{\mathbb{Z}}(M'', T)$.

4. View M as an abelian group. Then

$$0 \to M \to T$$

with T divisible. Apply $\hom_{\mathbb{Z}}(R, -)$ to get

$$0 \to \hom_{\mathbb{Z}}(R, M) \to \hom_{\mathbb{Z}}(R, T)$$

which is a sequence in R modules. The right most term is injective by the previous part. As

$$M = \hom_R(R, M) \hookrightarrow \hom_{\mathbb{Z}}(R, M) \hookrightarrow \hom_{\mathbb{Z}}(R, T)$$

this proves the theorem.

[4] Note the multiplcation occurs on the right!

$\square$

With this theorem in hand, if we have a diagram with the top line exact

$$\begin{tikzcd} 0 \arrow[r] & M' \arrow[r, "g"] \arrow[d, "f"] & M'' \\ & Q & \end{tikzcd}$$

We can put an exact sequence on the bottom with I injective

$$\begin{tikzcd} 0 \arrow[r] & M' \arrow[r, "g"] \arrow[d, "f"] & M'' & & \\ 0 \arrow[r] & Q \arrow[r, shift left, "i"] & I \arrow[l, shift left, bend left, "t"] \arrow[r] & N'' \arrow[r] & 0 \end{tikzcd}$$

we can get a diagram by lifting $i \circ f$

$$\begin{tikzcd} 0 \arrow[r] & M' \arrow[r, "g"] \arrow[d, "f"] & M'' \arrow[d, dotted, "h"] & & \\ 0 \arrow[r] & Q \arrow[r, shift left, "i"] & I \arrow[l, shift left, bend left, "t"] \arrow[r] & N'' \arrow[r] & 0 \end{tikzcd}$$

Giving us

$$\begin{tikzcd} 0 \arrow[r] & M' \arrow[r, "g"] \arrow[d, "f"] & M'' \arrow[d, dotted, "h"] \arrow[dl, dotted, "t \circ h"'] & & \\ 0 \arrow[r] & Q \arrow[r, shift left, "i"] & I \arrow[l, shift left, bend left, "t"] \arrow[r] & N'' \arrow[r] & 0 \end{tikzcd}$$

For $m \in M'$ we then have $i(f(m)) = h(g(m))$, so $f(m) = t(i(f(m))) = t(h(g(m)))$ which gives us the lift we need. $\square$

3 Homological Algebra

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While homological algebra was formally introduced in 1940, there are several earlier theorems that use the basic techniques of homological algebra. Some examples are Hilbert's 1890 Syzygy Theorem, Schur's 1904 work on projective representations of groups ($H^2(G, \mathcal{C}^\times)$), Schreier's 1926 work in group extensions ($H^2(G, A)$), and Noether's 1929 theory of cssed product algebras ($H^2(G, E)$).

While this area can be done in the general setting of abelian categories, we will restrict ourselves to R-modules.

3.1 Chain Complexes

Definition 27. A *chain (or descending) complex* $C_\bullet$ is an index set $\{C_i, d_i\}_{i\in\mathbb{Z}}$ of R-modules C_i with module morphisms $d_i : C_i \to C_{i-1}$ such that $d_{i-1} \circ d_i = 0$.

Example 21.

1. If M is an R-module, we can form a complex with $C_0 = M$ and everything else being the 0 object or map.
2. Given a short exact sequence in $R - \underline{mod}$

$$0 \to M' \xrightarrow{f} M \xrightarrow{g} M' \to 0$$

we can extend this to an exact chain by extending with the 0 object.

3. *The Projective Resolution of a Module*: Given a module M, M is the quotient of a projective module. $0 \to N_0 \to P_0 \to M \to 0$. Likewise, N_0 is the quotient of a projective module, so

$$0 \to N_1 \to P_1 \to N_0 \to 0$$

Continuing in this way we obtain a chain complex

$$\cdots \to P_2 \to P_1 \to P_0 \to M \to 0 \to \cdots$$

where the map $P_{i+1} \to P_i$ is given by $P_{i+1} \to N_i \to P_i$.

4. *Hilbert's Syzygy*[5] *Theorem*: If M is a module and X is a generating set for M, then

$$0 \to N_0 \to F_X = F_0 \to M \to 0$$

is an exact sequence. In this case, N_0 acts as relations amog the generators of M. If then Y is a generating set for N_0 we get another exact sequence

$$0 \to N_1 \to F_Y = F_1 \to N_0 \to 0$$

Continuitng we get our chain

$$\cdots \to F_2 \to F_1 \to F_0 \to M \to 0 \to \cdots$$

where $F_{i+1} \to F_i$ factors through N_i.

The Syzygy Theorem states if M is a module for $F[x_1, \ldots, x_n]$, then this free resolution of M has at most m steps. Thus, the relations among the relations among the relations etc. eventually become trivial.

3.2 Morphisms of Complexes

Definition 28. Given two complexes $C_\bullet = \{C_i, d_i\}$ and $C'_\bullet - \{C'_i, d'_i\}$ a *morphism of complexes* $\alpha : C_\bullet \to C'_\bullet$ consists of an indexed set of morphisms $\alpha_i : C_i \to C'_i$ such that each box commutes

$$\begin{tikzcd}
\cdots \arrow[r] & C_{i+1} \arrow[r, "d_{i+1}"] \arrow[d, "\alpha_{i+1}"] & C_i \arrow[r, "d_{i+1}"] \arrow[d, "\alpha_i"] & Ci-1 \arrow[r] \arrow[d, "\alpha_{i-1}"] & \cdots \\
\cdots \arrow[r] & C'_{i+1} \arrow[r, "d'_{i+1}"] & C'_i \arrow[r, "d'_{i+1}"] & C'i-1 \arrow[r] & \cdots
\end{tikzcd}$$

[5]The term syzygy comes from Hilbert's notation that the *syzygy* module of a module M is the module of relations between generators

This is usually written as $\alpha d = d'\alpha$.

This allows us to create $R - \underline{comp}$, the (abelian) category of chain complexes. In fact $R - \underline{mod}$ is a full subcategory of $R - \underline{comp}$ by identifying a module M with the chain complex consisting of M in one coordinate and 0 everywhere else.

If $C_\bullet, C'_\bullet$ are two complexes we can form the direct sum $C_\bullet \oplus C'_\bullet = \{C_i \oplus C'_i, d_i \oplus d'_i\}$. One can extend this to the direct sum of a family of complexes.

3.3 Homology

Let $C_\bullet$ be a complex

$$\cdots \to C_{i+1} \to C_i \to C_{i-1} \to \cdots$$

to measure exactness, we can look at $\ker d_i / \operatorname{im} d_{i+1}$. If the complex is exact at C_i, this module is 0.

Definition 29. The *cycle modules* are $Z_i = Z_i(C_\bullet) = \ker d_i$, and the *boundary modules* are $B_i = B_i(C_\bullet) = \operatorname{im} d_{i+1}$. The *homology modules* are $H_i = H_i(C_\bullet) = Z_i/B_i$.

Suppose $\alpha : C_\bullet \to C'_\bullet$ is a morphism of complexes

$$\begin{tikzcd}
\cdots \arrow[r] & C_{i+1} \arrow[r, "d_{i+1}"] \arrow[d, "\alpha_{i+1}"] & C_i \arrow[r, "d_i"] \arrow[d, "\alpha_i"] & C_{i-1} \arrow[r] \arrow[d, "\alpha_{i-1}"] & \cdots \\
\cdots \arrow[r] & C'_{i+1} \arrow[r, "d'_{i+1}"] & C'_i \arrow[r, "d'_i"] & C'_{i-1} \arrow[r] & \cdots
\end{tikzcd}$$

Then we see $\alpha(Z_i) \subset Z'_i$ and $\alpha_i(B_i) \subset B'_i$. We have induced maps

$$H_i(C_\bullet) = Z_i/B_i \xrightarrow{\tilde{\alpha}_i} Z'_i/B'_i = H_i(C'_\bullet)$$

by

$$z + B_i \mapsto \alpha_i(z) + B'_i$$

The well-definedness comes from the fact that $\alpha(Z_i) \subset Z'_i$ and $\alpha_i(B_i) \subset B'_i$.

As $\tilde{\alpha}_i \in \hom(H_i(C_\bullet), H_i(C'_\bullet))$, H_i is a covariant functor $H_i : R - \underline{comp} \to R - \underline{comp}$. Moreover, these functors are additive. Therefore $H_i(C_\bullet \oplus C'_\bullet) = H_i(C_\bullet) \oplus H_i(C'_\bullet)$.

3.4 Cohomology

Cohomology is defined in an analogous way to homology, except the arrows are reversed.

Definition 30. A *cochain complex (or ascending complex)* is a sequence of R-mods $C^\bullet = \{C^i, d^i\}$ where $d^i : C^i \to C^{i+1}$, $d^2 = 0$.

N.B. Note that any chain complex $C_\bullet$ can be made into a cochain complex $C^\bullet$ by $C_i \mapsto C^{-i}$.

Morphisms of a cochain complex are sequences of R-morphism $\alpha_i : C^i \to C'^i$ making the following diagram commute

$$\begin{tikzcd}
\cdots \arrow[r] & C^{i-1} \arrow[r, "d^{i-1}"] \arrow[d, "\alpha_i"] & C^i \arrow[r, "d^i"] \arrow[d, "\alpha_i"] & C^{i+1} \arrow[r, "d^{i+1}"] \arrow[d, "\alpha_{i+1}"] & \cdots \\
\cdots \arrow[r] & C'^{i-1} \arrow[r, "d'^{i-1}"] & C'^i \arrow[r, "d'^i"] & C'^{i+1} \arrow[r, "d'^{i+1}"] & \cdots
\end{tikzcd}$$

Then we define homology by $Z^i(C_\bullet) = \ker d^i$, $B^i(C_\bullet) = \operatorname{im} d^{i-1}$ and $H^i(C_\bullet) = Z^i/B^i$.

Then in an analogous way to homology, we see morphisms on cochain complexes induce a morphism on cohomology making H^i covariant, additive functors $R - \underline{cocomplex} \to R - \underline{mod}$.

3.5 Exact sequences of Complexes

Because cochain complexes and chain complexes are virtually the same, we will just focus on chain complexes and homology.

We can define the notion of a (short) exact sequence of chain complexes. Given chain complexes $A_\bullet, B_\bullet, C_\bullet$ with morphisms $\alpha : A_\bullet \to B_\bullet$ and $\beta : B_\bullet \to C_\bullet$, the commuting diagram

$$\begin{tikzcd}
\cdots \arrow[r] & 0 \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & A_{i+1} \arrow[r, "d_A"] \arrow[d, "\alpha"] & A_i \arrow[r] \arrow[d] & A_{i-1} \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & B_{i+1} \arrow[r, "d_B"] \arrow[d, "\beta"] & B_i \arrow[r] \arrow[d] & B_{i-1} \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & C_{i+1} \arrow[r, "d_C"] \arrow[d] & C_i \arrow[r] \arrow[d] & C_{i-1} \arrow[r] \arrow[d] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & \cdots
\end{tikzcd}$$

is exact if each of the columns are. From this, we get induced maps on the homology

$$H_i(A) \to H_i(B) \to H_i(C)$$

In fact, using a connecting morphism (which we will develop below) we actually can convert this to a long exact sequence of homology

$$\cdots \to H_{i+1}(A) \to H_{i+1}(B) \to H_{i+1}(C) \xrightarrow{\delta} H_i(A) \to \cdots$$

3.6 The Snake Lemma

In order to get the connecting morphism, we first develop the following lemma by way of a diagram chase.

Lemma 3 (Snake Lemma)**.** *Suppose we are given a diagram*

$$\begin{tikzcd}
 & M' \arrow[r] \arrow[d] & M \arrow[r] \arrow[d] & M'' \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & N' \arrow[r] & N \arrow[r] & N''
\end{tikzcd}$$

with exact rows.

Then we get an induced diagram

$$
\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
 & \ker d' \arrow[r] \arrow[d] & \ker d, \arrow[r] \arrow[d] & \ker d'' \arrow[d, "i"] \arrow[dddll, rounded corners] & \\
 & M' \arrow[r, "f"] \arrow[d, "d'"] & M \arrow[r, "g"] \arrow[d, "d"] & M'' \arrow[r] \arrow[d, "d''"] & 0 \\
0 \arrow[r] \arrow[dd, rounded corners] & N' \arrow[r, "h"] \arrow[d, "\pi"] \arrow[uu, rounded corners, to path={-- ([xshift=2ex]\tikztostart.east) |- ([yshift=-2ex]\tikztotarget.south) -- (\tikztotarget)}, "\delta"] & N \arrow[r, "k"] \arrow[d] & N'' \arrow[d] & \\
 & \operatorname{coker} d' \arrow[r] \arrow[d] & \operatorname{coker} d \arrow[r] \arrow[d] & \operatorname{coker} d'' \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}
$$

So that the sequence

$$\cdots \to \ker d' \to \ker d \to \ker d'' \xrightarrow{\delta} \operatorname{coker} d' \to \operatorname{coker} d \to \operatorname{coker} d'' \to \cdots$$

is exact.

Proof. For $x \in \ker d''$, then choose $y \in M$ so that $g(y) = i(x)$ [this is possible as g surjects onto M'']. Note that $k(d(y)) = d''(g(y)) = d''(i(x)) = 0$. Therefore $d(y) \in \operatorname{im} h$, so we can define $\delta(x) = \pi(h^{-1}(d(y)))$.

It must be checked that this is well-defined. The only point of contention is in the choice of y. If y' is another choice, then $y - y' \in \ker g$, so $f^{-1}(y - y') \in M'$, and we see that $\pi(h^{-1}(d(y - y'))) = \pi(f^{-1}(y - y')) = 0$, so the δ defined based on y or y' is the same, whence δ is well-defined. This is also easily seen to be a homomorphism, as $g(a + b) = g(a) + g(b)$, so if we choose $a \in g^{-1}(x)$ and $b \in g^{-1}(y)$ to define $\delta(x), \delta(y)$ we can choose $ra + b$ to define $\delta(rx + y)$. As the rest of the definition of δ is the composition of morphisms, we have that δ is a morphism. $\square$

3.7 Long Sequence of Homology

We can now apply the Snake Lemma in the setting of homology.

By the definitions of cycles and boundaries we get an exact sequence of complexes

$$
\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
 & Z'_i \arrow[r, "\alpha"] \arrow[d] & Z_i \arrow[r, "\beta"] \arrow[d] & Z''_i \arrow[d, "i"] \arrow[dd, bend left=60] & \\
 & C'_i \arrow[r, "\alpha_i"] \arrow[d, "d'_i"] & C_i \arrow[r, "\beta_i"] \arrow[d, "d_i"] & C''_i \arrow[r] \arrow[d, "d''_i"] & 0 \\
0 \arrow[r] \arrow[dd, bend right=60] & C'_{i-1} \arrow[r, "\alpha_{i-1}"] \arrow[d, "\pi"] & C_{i-1} \arrow[r, "\beta_{i-1}"] \arrow[d] \arrow[ll, bend right=60, "\delta"'] & C''_{i-1} \arrow[d] & \\
 & C'_{i-1}/B'_{i-1} \arrow[r, "\alpha_{i-1}"] \arrow[d] & C_{i-1}/B_{i-1} \arrow[r, "\beta_{i-1}"] \arrow[d] & C''_{i-1}/B''_{i-1} \arrow[d] & \\
 & 0 & 0 & 0 &
\end{tikzcd}
$$

By the Snake lemma we get a map $\delta : Z''_i \to C'_{i-1}/B'_{i-1}$. We now need to show that this map can factor through homology so that we can map $Z''_i/B''_i \to Z'_{i-1}/B'_{i-1}$ by $z/B''_i \mapsto \delta(z)$. In order for this to be well-defined, we must show that $\delta(B''_i) = 0$ and $\operatorname{im}\delta \subset Z'_{i-1}/B'_{i-1}$.

To check the first statement, if $x \in B''_i$, then we can write $x = d''(\xi)$ for some $\xi \in C''_{i+1}$. Then definition of δ leaves us with finding $y \in C_i$ with $\beta_i(y) = x$. But our choice can then be $d(y')$ where $y' \in \beta_{i+1}^{-1}(\xi)$. Then the next step of the Snake lemma sends $d(y') \mapsto d(d(y')) = 0$, so in the end $\delta(x) = 0$.

Now we check the last statement. Say $x \in Z''_i$ then for $y \in \beta_i^{-1}(x)$ we have $\delta(x) = \alpha_{i-1}^{-1}(d_i(y))$, so $d'(\delta(x)) = d'(\alpha_{i-1}^{-1}(d_i(y)))$. We claim this is 0. As α_{i-2} is injective, it suffices to show $\alpha_{i-2}(d'(\alpha_{i-1}^{-1}(d_i(y)))) = 0$. By the commutativity of the diagram

$$\alpha_{i-2}(d'(\alpha_{i-1}^{-1}(d_i(y)))) = d_{i-1}(\alpha_{i-1}(\alpha_{i-1}^{-1}(d_i(y)))) = d^2(y) = 0$$

Therefore δ does induce a map on homology. We are left to check exactness of this sequence.

Thus, given an exact sequence of complexes

$$0 \to C'_\bullet \xrightarrow{\alpha} C_\bullet \xrightarrow{\beta} C''_\bullet \to 0$$

the Snake Lemma gies a connecting morphism

$$\delta_i : H_i(C''_\bullet) \to H_{i-1}(C'_\bullet)$$

so that

$$\cdots \to H_i(C'_\bullet) \to H_i(C_\bullet) \to H_i(C''_\bullet) \xrightarrow{\delta_i} H_{i-1}(C'_\bullet) \to \cdots$$

is an exact sequence.

Now if we have a commutative diagram with exact rows

$$
\begin{tikzcd}
0 \arrow[r] & C'_\bullet \arrow[r, "\alpha"] \arrow[d, "\alpha"] & C_\bullet \arrow[r, "\beta"] \arrow[d, "f"] & C''_\bullet \arrow[r] \arrow[d, "f''"] & 0 \\
0 \arrow[r] & D'_\bullet \arrow[r, "\gamma"] & D_\bullet \arrow[r, "\kappa"] & D''_\bullet \arrow[r] & 0
\end{tikzcd}
$$

then for each i, we have the following commutative square

$$\begin{tikzcd} H_i(C''_\bullet) \arrow[r, "\delta_i^c"] \arrow[d, "\tilde{f}''_i"] & H_{i-1}(C'_\bullet) \arrow[d, "\tilde{f}'_{i-1}"] \\ H_i(D''_\bullet) \arrow[r, "\delta_i^D"] & H_{i-1}(D'_\bullet) \end{tikzcd}$$

Likewise, we can dualize the snake lemma to get a connecting homomorphism on cohomology.

$$\delta_i : H_i(C''_\bullet) \to H_{i+1}(C'_\bullet)$$

so that

$$\cdots \to H_i(C'_\bullet) \to H_i(C_\bullet) \to H_i(C''_\bullet) \xrightarrow{\delta_i} H_{i+1}(C'_\bullet) \to \cdots$$

is an exact sequence.

Now if we have a commutative diagram with exact rows

$$\begin{tikzcd} 0 \arrow[r] & C'_\bullet \arrow[r, "\alpha"] \arrow[d, "\alpha"] & C_\bullet \arrow[r, "\beta"] \arrow[d, "f"] & C''_\bullet \arrow[r] \arrow[d, "f''"] & 0 \\ 0 \arrow[r] & D'_\bullet \arrow[r, "\gamma"] & D_\bullet \arrow[r, "\kappa"] & D''_\bullet \arrow[r] & 0 \end{tikzcd}$$

then for each i, we have the following commutative square

$$\begin{tikzcd} H_i(C''_\bullet) \arrow[r, "\delta_i^c"] \arrow[d, "\tilde{f}''_i"] & H_{i+1}(C'_\bullet) \arrow[d, "\tilde{f}'_{i+1}"] \\ H_i(D''_\bullet) \arrow[r, "\delta_i^D"] & H_{i+1}(D'_\bullet) \end{tikzcd}$$

3.8 Homotopies

Given two complexes $C_\bullet, C'_\bullet$ and a morphism $\alpha : C_\bullet \to C'_\bullet$ we get an induced map on homology

$$\tilde{\alpha}_i : H_i(C) \to H_i(C')$$

When do two maps $\alpha, \beta : C_\bullet \to C'_\bullet$ induce the same map on homology? One sufficient condition is that the maps are *homotopic*.

Definition 31. Let $\alpha, \beta : C_\bullet \to C'_\bullet$ be morphisms of complexes. We say α is *homotopic* to β ($\alpha \sim \beta$) if there is a sequence of morphisms $\{si\}$ where $s_i : C_i \to C'_{i+1}$ such that $\alpha_i - \beta_i = d'_{i+1}s_i + s_{i-1}d_i$

$$\begin{tikzcd} \cdots \arrow[r] & C_{i+1} \arrow[r, "d_{i+1}"] \arrow[d, "\beta_{i+1}"', "\alpha_{i+1}"] & C_i \arrow[r, "d_i"] \arrow[d, "\beta_i"', "\alpha_i"] \arrow[dl, "s_i"] & C_{i-1} \arrow[r] \arrow[d, "\beta_{i-1}"', "\alpha_{i-1}"] \arrow[dl, "s_{i-1}"] & \cdots \\ \cdots \arrow[r] & C'_{i+1} \arrow[r, "d'_{i+1}"'] & C'_i \arrow[r, "d'_i"'] & C'_{i-1} \arrow[r] & \cdots \end{tikzcd}$$

Proposition 11. *If $\alpha, \beta : C_\bullet \to C'_\bullet$ and $\alpha \sim \beta$, then $\tilde{\alpha}_i = \tilde{\beta}_i$ for all i.*

Proof. Recall that the induced map on homology is given by $\tilde{\alpha}_i(z_i + B_i) = \alpha_i(z_i) + B'_i$ and $\tilde{\beta}_i(z_i + B'_i) = \beta_i(z_i) + B'_i$.

Thus, the maps are the same if $\beta_i(z_i) - \alpha_i(z_i) \in B'_i$. But by the homotopy,

$$\beta_i(z_i) - \alpha_i(z_i) = d'_{i+1}s_i(z_i) + s_{i-1}d_i(z_i) = d'_{i+1}s_i(z_i) \in B'_i$$

□

Proposition 12. *Homotopy is an equivalence relation on morphisms of complexes*

Proof. This is clearly reflexive and symmetric. For transitivity, just add the morphisms. □

Proposition 13. *Homotopies can be multiplied (by composition), i.e., if $\alpha, \beta : C_\bullet \to C'_\bullet$ are homotopic and $\gamma, \delta : C'_\bullet \to C''_\bullet$ are homotopic, then $\gamma \circ \alpha \sim \delta \circ \beta$.*

Proof. Let s be the homotopy between $\alpha \sim \beta$ and t between $\gamma \sim \delta$.

We have the diagram

$$\begin{tikzcd}
\cdots \arrow[r] & C_{i+1} \arrow[r, "d_{i+1}"] \arrow[d] & C_i \arrow[r, "d_i"] \arrow[d, "\alpha_i"] \arrow[dl, "s_i"', "\beta_i"] & C_{i-1} \arrow[r] \arrow[d] \arrow[dl, "s_{i-1}"'] & \cdots \\
\cdots \arrow[r] & C'_{i+1} \arrow[r, "d'_{i+1}"'] \arrow[d] & C'_i \arrow[r, "d'_i"'] \arrow[d, "\gamma_i"] \arrow[dl, "t_i"', "\delta i"] & C_{i-1} \arrow[r] \arrow[d] \arrow[dl, "t_{i-1}"'] & \cdots \\
\cdots \arrow[r] & C''_{i+1} \arrow[r, "d'_{i+1}"'] & C''_i \arrow[r, "d'_i"'] & C''_{i-1} \arrow[r] & \cdots
\end{tikzcd}$$

Now consider $\gamma_i(\alpha_i - \beta_i) = \gamma_i\alpha_i - \gamma_i\beta_i$ and $(\gamma_i - \delta_i)\beta_i = \gamma_i\beta_i - \delta_i\beta_i$. Then

$$\gamma_i(\alpha_i - \beta_i) = \gamma_i d'_{i+1} s_i + \gamma_i s_{i-1} d_i = d''_{i+1}\gamma_{i+1} s_i + \gamma_i s_{i-1} d_i$$

and

$$(\gamma_i - \delta_i)\beta_i = d''_{i+1} t_i \beta_i + t_{i-1} d'_i \beta_i$$

adding these yields

$$\gamma_i\alpha_i - \delta_i\beta_i = d''_{i+1}(\gamma_{i+1}s_i + t_i\beta_i) + (\gamma_i s_{i-1} + t_{i-1}\beta_{i-1})d_i$$

so our homotopies should be given as $\gamma_{i+1}s_i + t_i\beta_i$. □

3.9 Resolutions

"Oh I am not happy"

Cogdell

Begin with an R-module M. Then a *resolution* of M is a non-negative[6] complex $C_\bullet$ and an augmentation morphism $C_0 \xrightarrow{\epsilon} M \to 0$ so that

$$\to C_2 \to C_1 \to C_0 \xrightarrow{\epsilon} M \to 0 \to 0$$

is exact.

Definition 32. A resolution is *projective* if all P_i are projective for $i \geq 0$.

We have construted these projective resolutions before but we've never checked if they were exact. Recall the construction is given by first expressing a module as a quotient of a projective

$$0 \to N_0 \to P_0 \xrightarrow{\epsilon} M \to 0$$

then doing the same for N_i

$$0 \to N_{i+1} \to P_{i+1} \to N_i \to 0$$

and we create the sequence

[6] This means $C_{-i} = 0$ for i positive

$$\begin{tikzcd}
\cdots \arrow[r] & P_3 \arrow[r, "d_3"] & P_2 \arrow[rr, "d_2"] \arrow[dr] & & P_1 \arrow[rr, "d_1"] \arrow[dr] & & P_0 \arrow[r, "\epsilon"] & M \arrow[r] & 0 \\
& & & N_1 \arrow[ur] & & N_0 \arrow[ur] & & &
\end{tikzcd}$$

We check that this is an exact sequence, which follows easily from the exactness from the short exact sequences.

While we have the existence of projective and injective resolutions, we need some results about their uniqueness.

Theorem 5. *Let $(P_\bullet, \epsilon)$ be a projective resolution of M, and $(C'_\bullet, \epsilon')$ be any resolution of M'. Let $\mu : M \to M'$ be a module morphism. Then there is a morphism $\alpha : (P_\bullet, \epsilon) \to (C'_\bullet, \epsilon')$ so that $\mu\epsilon = \epsilon'\alpha_0$, i.e. the square containing the augmentations commute.*

$$\begin{tikzcd}
\arrow[r] & P_0 \arrow[r, "\epsilon"] \arrow[d, "\alpha_0"] & M \arrow[r] \arrow[d, "\mu"] & 0 \\
\arrow[r] & C'_0 \arrow[r, "\epsilon'"] & M' \arrow[r] & 0
\end{tikzcd}$$

Moreover, any two such morphisms $\alpha, \beta : (P_\bullet, \epsilon) \to (C'_\bullet, \epsilon')$ satisfying these conditions are homotopic.

Proof. The first assertion comes by way of induction: First, using the projectivity of P_0 we get a morphism $\alpha_0 : P_0 \to C'_0$ so that $\epsilon' \circ \alpha_0 = \mu \circ \epsilon$.

$$\begin{tikzcd}
\cdots \arrow[r] & P_2 \arrow[r, "d_2"] & P_1 \arrow[r, "d_1"] & P_0 \arrow[r, "\epsilon"] \arrow[d, "\alpha_0"] & M \arrow[r] \arrow[d, "\mu"] & \cdots \\
\cdots \arrow[r] & C'_2 \arrow[r, "d'_2"] & C'_1 \arrow[r, "d'_1"] & C'_0 \arrow[r, "\epsilon'"] & M' \arrow[r] & \cdots
\end{tikzcd}$$

Now for the induction step, suppose we have constructed $\alpha_{n-1}, \cdots, \alpha_0$ so that the requisite commutivity holds.

$$\begin{tikzcd}
P_n \arrow[r, "d_n"] & P_{n-1} \arrow[r, "d_{n-1}"] \arrow[d, "\alpha_{n-1}"] & P_{n-2} \arrow[r, "d_{n-2}"] \arrow[d, "\alpha_{n-2}"] & \cdots \arrow[r] & P_2 \arrow[r, "d_2"] \arrow[d, "\alpha_2"] & P_1 \arrow[r, "d_1"] \arrow[d, "\alpha_1"] & P_0 \arrow[r, "\epsilon"] \arrow[d, "\alpha_0"] & M \arrow[r] \arrow[d, "\mu"] & \cdots \\
C'_n \arrow[r, "d'_n"] & C'_{n-1} \arrow[r, "d'_{n-1}"] & C'_{n-2} \arrow[r, "d'_{n-2}"] & \cdots \arrow[r] & C'_2 \arrow[r, "d'_2"] & C'_1 \arrow[r, "d'_1"] & C'_0 \arrow[r, "\epsilon'"] & M' \arrow[r] & \cdots
\end{tikzcd}$$

Note that $d'_{n-1} \circ \alpha_{n-1} \circ d_n = \alpha_{n-2} \circ d_{n-1} \circ d_n$. Thus, $\operatorname{im} \alpha_{n-1} \circ d_n \subseteq \operatorname{im} d'_n$. Thus, we can restrict C'_{n-1} to $\operatorname{im} C'_n$ and lift the morphism $\alpha_{n-1} \circ d_n : P_n \to C'_{n-1}$ using the projectivity of P_n to C'_n, and this makes the diagram commute.

$$\begin{tikzcd}
P_n \arrow[r] \arrow[d, "\alpha_n"] & P_{n-1} \arrow[r] \arrow[d, "\alpha_{n-1}"] & P_{n-2} \arrow[r] \arrow[d, "\alpha_{n-2}"] & \cdots \arrow[r] & P_2 \arrow[r, "d_2"] \arrow[d, "\alpha_2"] & P_1 \arrow[r, "d_1"] \arrow[d, "\alpha_1"] & P_0 \arrow[r, "\epsilon"] \arrow[d, "\alpha_0"] & M \arrow[r] \arrow[d, "\mu"] & \cdots \\
C'_n \arrow[r] & C'_{n-1} \arrow[r] & C'_{n-2} \arrow[r] & \cdots \arrow[r] & C'_2 \arrow[r, "d'_2"] & C'_1 \arrow[r, "d'_1"] & C'_0 \arrow[r, "\epsilon'"] & M' \arrow[r] & \cdots
\end{tikzcd}$$

By induction we thus get our morphism α.

Now we show this morphism is unique up to homotopy which we now construct. Define $\gamma = \alpha - \beta$. Then it suffices to construct a homotopy $\gamma \sim 0$. We again use induction. Note that $\epsilon'\gamma_0 = 0$ as α, β both satisfy commutivity with the augmentation map. In the base case we have the diagram

$$\begin{tikzcd}
 & P_0 \arrow[d, "\gamma_0"] & & \\
C'_1 \arrow[r, "d'_1"] & C'_0 \arrow[r, "\epsilon'"] & M' \arrow[r] & 0
\end{tikzcd}$$

Again, as $\operatorname{im}\gamma_0 \subseteq \ker\epsilon'$, we can restrict C'_0 to $\operatorname{im} d'_1$ and lift γ_0 restricted to this codomain to get $s_0 : P_0 \to C'_1$. This makes $d'_1 \circ s_0 = \gamma_0$.

Now by induction if we have s_{n-1} we get

$$\begin{tikzcd}
P_n \arrow[r] \arrow[d, "\gamma n"] & P_{n-1} \arrow[r] \arrow[d, "\gamma n-1"] \arrow[dl, "s_{n-1}"] & P_{n-2} \arrow[r] \arrow[d, "\gamma_{n-2}"] \arrow[dl, "s_{n-2}"] & \cdots \arrow[r] & P_2 \arrow[r, "d_2"] \arrow[d, "\gamma_2"] & P_1 \arrow[r, "d_1"] \arrow[d, "\gamma_1"] & P_0 \arrow[r, "\epsilon"] \arrow[d, "\gamma_0"] & M \arrow[r] \arrow[d, "\mu"] & \cdots \\
C'_n \arrow[r] & C'_{n-1} \arrow[r] & C'_{n-2} \arrow[r] & \cdots \arrow[r] & C'_2 \arrow[r, "d'_2"] & C'_1 \arrow[r, "d'_1"] & C'_0 \arrow[r, "\epsilon'"] & M' \arrow[r] & \cdots
\end{tikzcd}$$

so that $d'_{i+1}s_i = s_{i-1}d_i$. Consider the map $j : P_n \to C'_n$ by $x \mapsto \gamma_n(x) - s_{n-1}(d_n(x))$. Again, we can see $\operatorname{im} j \subseteq \operatorname{im} d'_{n+1} = \ker d'_n$ as

$$d'_n(\gamma_n - s_{n-1}d_n) = d'_n\gamma_n - d'_n s_{n-1}d_n = \gamma_{n-1}d_n - d'_n s_{n-1}d_n = (\gamma_{n-1} - d'_n s_{n-1})d_n = s_{n-2}d_{n-1}d_n = 0$$

so we can restrict j to the codomain $\operatorname{im} d'_{i+1}$ and then lift j to $s_n : P_n \to C'_{n+1}$. The map s_n then satisfies

$$d'_{n+1} \circ s_n = \gamma_n - s_{n-1}d_n$$

which is the condition we need for a homotopy. □

Corollary 1. *Suppose we have two projective resolutions $(P_\bullet, \epsilon)$ and $(P'_\bullet, \epsilon')$ for a given module M. Then there is a morphism $\alpha : P_\bullet \to P'_\bullet$ lifting id_M. This is unique up to homotopy. Now if $\beta : P'_\bullet \to P_\bullet$ lifts id_M, then α, β are inverses up to homotopy.*

Proof. Consider

$$\cdots \longrightarrow P_1 \longrightarrow P_0 \longrightarrow P \longrightarrow 0$$

$$\cdots \longrightarrow P'_1 \longrightarrow P'_0 \longrightarrow P' \longrightarrow 0$$

$$\cdots \longrightarrow P_1 \longrightarrow P_0 \longrightarrow P \longrightarrow 0$$

$$\cdots \longrightarrow P'_1 \longrightarrow P'_0 \longrightarrow P' \longrightarrow 0$$

□

Definition 33. A morphism $\alpha : C_\bullet \to C'_\bullet$ is an *quasi-isomorphism* if the induced maps

$$\tilde{\alpha} : H_i(C_\bullet) \to H_i(C'_\bullet)$$

on homology are isomorphisms.

Now consider the resolutions

$$\begin{tikzcd}
\cdots \arrow[r] & P_2 \arrow[r] & P_1 \arrow[r] & P_0 \arrow[r] \arrow[d, "\epsilon"] & 0 \arrow[r] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & M \arrow[r] & 0 \arrow[r] & \cdots
\end{tikzcd}$$

This is a quasi-isomorphism, as $H_i(P_\bullet) = 0$ for $i \geq 1$ and $H_0(P_\bullet) = \ker(0)/\operatorname{im} d_1 \cong M$.

3.10 Injective Resolutions

Definition 34. If M is an R-module, a *corresolution* of M is a non-negative cochain complex $C^\bullet$ along with a *coaugmentation morphism* $\eta : M \to C^0$ such that

$$\to 0 \to M \xrightarrow{\eta} C^0 \xrightarrow{d^0} C^1 \xrightarrow{d^1} \cdots$$

is exact.

In the case all C^i are injective, this is called an *injective resolution.*

As R-<u>mod</u> has enough injectives, we can construct an injective resolution for any module M: take

$$0 \to M \to Q^0 \to N^0 \to 0$$

with Q^0 injective, and continue with

$$0 \to N^0 \to Q^1 \to N^1 \to 0$$

and combine them into a cochain complex.

CHANGE

$$\begin{tikzcd}
\cdots \arrow[r] & M \arrow[r, "d_0"] & Q^0 \arrow[rr, "d^0"] \arrow[dr] & & Q^1 \arrow[rr, "d_1"] \arrow[dr] & & Q^0 \arrow[r, "\epsilon"] & M \arrow[r] & 0 \\
& & & N^1 \arrow[ur] & & N^0 \arrow[ur] & & &
\end{tikzcd}$$

The analogues of the projective resolutions follow by way of duality for injective resolutions.

Theorem 6. *Let $(Q^\bullet, \eta)$ be an injective resolution of M and $(C'^\bullet, \eta')$ any co-resolution of M'. Then if $\lambda' : M' \to M$ is any module morphism, there is a morphism $\beta : C'^\bullet \to Q^\bullet$ such that $\beta^0 \circ \eta' = \eta \circ \lambda'$. Moreover, any two such morphisms β, β' are homotopic.*

$$\begin{tikzcd}
0 \arrow[r] & M' \arrow[r, "\eta'"] \arrow[d, "\lambda'"] & C'^0 \arrow[r, "d'^0"] \arrow[d, "\beta^0"] & C'^1 \arrow[r, "d'^1"] \arrow[d, "\beta^1"] & C'^2 \arrow[r] \arrow[d, "\beta^2"] & \cdots \\
0 \arrow[r] & M \arrow[r, "\eta"] & Q^0 \arrow[r] & Q^1 \arrow[r] & Q^2 \arrow[r] & \cdots
\end{tikzcd}$$

Theorem 7. *If $(Q^\bullet, \eta)$ and $(I^\bullet, \eta')$ are two injective resolutions of M. Then there is a morphism $\beta : I^\bullet \to Q^\bullet$ which lifts id_M, and this lift is unique up to homotopy.*

This basically says the injective resolutions of a module are unique up to homotopy.

N.B. These injective resolutions are used to define either right derived functors for covariant functors or left derived functors for contravariant functors.

N.B. This ends the material for the first exam.

3.11 Derived Functors

Suppose $F : R - \underline{mod} \to \mathbb{Z} - \underline{mod}$ is a covariant functor. For the most part we will deal with $F = \hom(M, -)$ which will give us Ext, and $F = M \otimes -$ which will give us Tor. We also assume that F is additive.

The idea is as follows: take a module M and replace it with a projective resolution $(P_\bullet, \epsilon)$ of M.

$$\cdots \longrightarrow P_2 \longrightarrow P_1 \longrightarrow P_0 \xrightarrow{\epsilon} M \longrightarrow 0$$

Then applying the functor F yields.

$$\cdots \longrightarrow F(P_2) \longrightarrow F(P_1) \longrightarrow F(P_0) \xrightarrow{F(\epsilon)} F(M) \longrightarrow 0$$

This remains a complex.

In the case F is exact, this remains a resolution of $F(M)$. We can compute its homology H_i. As F is exact, $H_i = 0$. If F is not exact, the homology becomes more interesting. This is how we get the derived functors.

Definition 35. The *left derived functor* $L_nF(M) = H_n(F(P_\bullet))$ for $n \geq 0$.

In particular, we have $L_0(F(M)) = \operatorname{coker}(F(d_1)) = F(P_0)/F(d_1)F(P_n) = F(P_0)/B_1(F(P_\bullet))$.

Recall that as projective resolutions are unique up to homotopy and therefore homology, these derived functors are well-defined up to isomoorphism.

How does this functor act on morphisms? Suppose $\mu : M \to M'$ is a morphism. We need a map $L_nF(\mu) : L_n(F(M)) \to L_n(F(M'))$. By our above work, if we have a morphism μ and projective resolutions of M, M', this induces a map of complexes $\alpha : (P_\bullet, \epsilon) \to (P'_\bullet, \epsilon')$.

$$\begin{tikzcd}
\cdots \arrow[r] & P_2 \arrow[r] \arrow[d, "\alpha_2"] & P_1 \arrow[r] \arrow[d, "\alpha_1"] & P_0 \arrow[r] \arrow[d, "\alpha_0"] & M \arrow[r] \arrow[d, "\mu"] & 0 \\
\cdots \arrow[r] & P'_2 \arrow[r] & P'_1 \arrow[r] & P'_0 \arrow[r] & M' \arrow[r] & 0
\end{tikzcd}$$

We can then apply F to all of this to get

$$\begin{tikzcd}
\cdots \arrow[r] & F(P_2) \arrow[r] \arrow[d, "F(\alpha_2)"] & F(P_1) \arrow[r] \arrow[d, "F(\alpha_1)"] & F(P_0) \arrow[r] \arrow[d, "F(\alpha_0)"] & F(M) \arrow[r] \arrow[d, "F(\mu)"] & 0 \\
\cdots \arrow[r] & F(P'_2) \arrow[r] & F(P'_1) \arrow[r] & F(P'_0) \arrow[r] & F(M') \arrow[r] & 0
\end{tikzcd}$$

Note that we still have $F(\epsilon') \circ F(\alpha_0) = F(\mu) \circ F(\epsilon)$ and moreover all the other squares commute. Thus we have induced maps on homology. These induced maps are $L_n(\mu) = \widetilde{F(\alpha_n)}$. As our lifts α are unique up to homotopy, these functions are well-defined up to isomorphism, as F respects this homotopy.

3.12 Connecting Homomorphisms

We have that the left derived functors are homology functors for projective resolutions. Similarly, given a short exact sequence of complexes

$$0 \longrightarrow C'_\bullet \longrightarrow C_\bullet \longrightarrow C''_\bullet \longrightarrow 0$$

we get a long exact sequence of homology functors by the Snake Lemma.

Does this induce a long exact sequence of derived functors when given an exact sequence of modules

$$0 \longrightarrow M' \xrightarrow{\alpha} M \xrightarrow{\beta} M'' \longrightarrow 0$$

The only way this could happen is if when we replace the modules with their projective resolutions we get a short exact sequence of resolutions.

Definition 36. By a *projective resolution of a short exact sequence of modules*

$$0 \longrightarrow M' \xrightarrow{\alpha} M \xrightarrow{\beta} M'' \longrightarrow 0$$

we mean projective resolution of each module $(P'_\bullet, \epsilon')$ of M', $(P_\bullet, \epsilon)$ of M and $(P''_\bullet, \epsilon'')$ of M'' together with chain morphisms $i : P'_\bullet \to P_\bullet$ and $\pi : P_\bullet \to P''_\bullet$ so that

$$0 \longrightarrow P'_\bullet \xrightarrow{i} P_\bullet \xrightarrow{\pi} P''_\bullet \longrightarrow 0$$

is an exact sequence of complexes and so that the following diagram commutes:

$$\begin{tikzcd}
0 \arrow[r] & P'_0 \arrow[r, "i_0"] \arrow[d, "\epsilon'"] & P_0 \arrow[r, "\pi_0"] \arrow[d, "\epsilon"] & P''_0 \arrow[r] \arrow[d, "\epsilon''"] & 0 \\
0 \arrow[r] & M' \arrow[r, "\alpha"] \arrow[d] & M \arrow[r, "\beta"] \arrow[d] & M'' \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

For future reference, we will express this diagram as

$$\begin{tikzcd}
0 \arrow[r] & P'_\bullet \arrow[r, "i"] \arrow[d, "\epsilon'"] & P_\bullet \arrow[r, "\pi"] \arrow[d, "\epsilon"] & P''_\bullet \arrow[r] \arrow[d, "\epsilon''"] & 0 \\
0 \arrow[r] & M' \arrow[r, "\alpha"] \arrow[d] & M \arrow[r, "\beta"] \arrow[d] & M'' \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

Proposition 14. *Any short exact sequence of modules*

$$0 \longrightarrow M' \xrightarrow{\alpha} M \xrightarrow{\beta} M'' \longrightarrow 0$$

has a projective resolution.

Proof. We construct these chain morphisms inductively. Begin with projective presentations for each of M' and M''.

$$\begin{tikzcd}
 & 0 \arrow[d] & & 0 \arrow[d] & \\
 & N_0' \arrow[d] & & N_0'' \arrow[d] & \\
 & P_0' \arrow[d] & & P_0'' \arrow[d] & \\
0 \arrow[r] & M' \arrow[r] \arrow[d] & M \arrow[r] & M'' \arrow[r] \arrow[d] & 0 \\
 & 0 & & 0 &
\end{tikzcd}$$

Let $P_0 = P_0' \oplus P_0''$. This is projective to get

$$\begin{tikzcd}
 & 0 \arrow[d] & & 0 \arrow[d] & \\
 & N_0' \arrow[d] & & N_0'' \arrow[d] & \\
 & P_0' \arrow[r, "i_0"] \arrow[d] & P_0 \arrow[r, "\pi_0"] & P_0'' \arrow[d] & \\
0 \arrow[r] & M' \arrow[r] \arrow[d] & M \arrow[r] & M'' \arrow[r] \arrow[d] & 0 \\
 & 0 & & 0 &
\end{tikzcd}$$

with i_0, π_0 forming a split exact sequence. To get the augmentation map $\epsilon : P_0 \to M$ start by lifting $\epsilon'' : P_0'' \to M''$ to $\epsilon^* : P_0'' \to M$. Thus $\beta \circ \epsilon^* = \epsilon''$. Define $\epsilon : P_0 \to M$ by $\epsilon(x_0', x_0'') = \alpha\epsilon'(x_0') + \epsilon^*(x_0'')$. This diagram clearly commutes

$$\begin{tikzcd}
 & 0 \arrow[d] & & 0 \arrow[d] & \\
 & N_0' \arrow[d] & & N_0'' \arrow[d] & \\
 & P_0' \arrow[r, "i_0"] \arrow[d] & P_0 \arrow[r, "\pi_0"] & P_0'' \arrow[d] & \\
0 \arrow[r] & M' \arrow[r, "\alpha"] \arrow[d, "\epsilon'"] & M \arrow[r, "\beta"] \arrow[d, "\epsilon"] & M'' \arrow[r] \arrow[d, "\epsilon''"] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

Let $N_0 = \ker \epsilon$. So we get

$$
\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
 & N_0' \arrow[r, "i_0"] \arrow[d] & N_0 \arrow[r, "\pi_0"] & N_0'' \arrow[d] & \\
 & P_0' \arrow[r, "i_0"] \arrow[d] & P_0 \arrow[r, "\pi_0"] & P_0'' \arrow[d] & \\
0 \arrow[r] & M' \arrow[r, "\alpha"] \arrow[d, "\epsilon'"] & M \arrow[r, "\beta"] \arrow[d, "\epsilon"] & M'' \arrow[r] \arrow[d, "\epsilon''"] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}
$$

We want surjectivity of ϵ and injectivity $N_0 \to P_0$ exact. We can do the first by the Snake lemma, and the second by either the 5-lemma of the Snake Lemma.

Next, we obtain the square

$$
\begin{tikzcd}
0 \arrow[r] & N_1' \arrow[r, "i_1"] \arrow[d] & N_1 \arrow[r, "\pi_1"] \arrow[d] & N_1'' \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & P_1' \arrow[r, "i_1"] \arrow[d] & P_1 \arrow[r, "\pi_1"] \arrow[d] & P_1'' \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & N_0' \arrow[r, "i_0"] \arrow[d] & N_0 \arrow[r, "\pi_0"] \arrow[d] & N_0'' \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}
$$

by the same exact method. Thus we are left with the following chain complexes

$$
\begin{tikzcd}
0 \arrow[r] & P_1' \arrow[r] \arrow[d] \arrow[dd, bend right, "\partial_2'"] & P_1 \arrow[r] \arrow[d] \arrow[dd, bend right, "\partial_2'"] & P_1'' \arrow[r] \arrow[d] \arrow[dd, bend right, "\partial_2'"] & 0 \\
0 \arrow[r] & N_0' \arrow[r] \arrow[d] & N_0 \arrow[r] \arrow[d] & N_0'' \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & P_0' \arrow[r] \arrow[d] & P_0 \arrow[r] \arrow[d] & P_0'' \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & M' \arrow[r] \arrow[d] & M \arrow[r] \arrow[d] & M'' \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}
$$

which gives our projective resolutions of

$$
0 \longrightarrow M' \xrightarrow{\alpha} M \xrightarrow{\beta} M'' \longrightarrow 0
$$

□

Theorem 8. *Let F be a covariant additive functor from R-<u>mod</u> to $\mathbb{Z}$-<u>mod</u>, and let*

$$
0 \to M'' \xrightarrow{\alpha} M \xrightarrow{\beta} M'' \to 0
$$

Then for any $n \geq 1$ there is a conntecting homomorphism $\Delta_n : L_nF(M'') \to L_{n-1}F(M')$ such that we obtain a long exact sequence of left derived functors

$$\cdots \to L_n(F(M')) \to L_n(F(M)) \to L_n(F(M'')) \to L_{n-1}(F(M')) \to \cdots$$

Proof. We begin with projective resolution

$$\begin{tikzcd}
0 \arrow[r] & P'_\bullet \arrow[r, "i"] \arrow[d, "\epsilon'"] & P_\bullet \arrow[r, "\pi"] \arrow[d, "\epsilon"] & P''_\bullet \arrow[r] \arrow[d, "\epsilon''"] & 0 \\
0 \arrow[r] & M' \arrow[r, "\alpha"] \arrow[d] & M \arrow[r, "\beta"] \arrow[d] & M'' \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

For each $n \geq 0$ we have a short exact sequence

$$0 \longrightarrow P'_n \longrightarrow P_n \longrightarrow P''_n \longrightarrow 0$$

As P''_n is projective, we can view $P_n \cong P'_n \oplus P''_n$. Now applying F we see $F(P_n) \cong F(P'_n) \oplus F(P''_n)$ by the additivity of F. Therefore the sequence

$$0 \longrightarrow F(P'_n) \longrightarrow F(P_n) \longrightarrow F(P''_n) \longrightarrow 0$$

remains short exact.

$$\begin{tikzcd}
0 \arrow[r] & F(P'_\bullet) \arrow[r, "F(i)"] \arrow[d, "F(\epsilon)'"] & F(P_\bullet) \arrow[r, "F(\pi)"] \arrow[d, "F(\epsilon)"] & F(P''_\bullet) \arrow[r] \arrow[d, "F(\epsilon'')"] & 0 \\
0 \arrow[r] & F(M') \arrow[r, "F(\alpha)"] \arrow[d] & M \arrow[r, "F(\beta)"] \arrow[d] & F(M'') \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

is still a short exact sequence of complexes. Applying the homology functors, we get a connecting homomorphism $\delta_n : H_n(P''_\bullet) \to H_{n-1}(P'_\bullet)$ creating the long exact sequence of homology

$$\cdots \longrightarrow H_n(F(P'_\bullet)) \longrightarrow H_n(F(P_\bullet)) \longrightarrow H_n(F(P''_\bullet)) \longrightarrow H_{n-1}(F(P'_\bullet)) \longrightarrow \cdots$$

and this by definition is

$$\cdots \longrightarrow L_n(F(M')) \longrightarrow L_n(F(M)) \longrightarrow L_n(F(M'')) \longrightarrow L_{n-1}(F(M')) \longrightarrow \cdots$$

□

3.12.1 Right Derived Functors for Covariant Functors

Let $F : R - \underline{mod} \to \mathbb{Z} - \underline{mod}$ be an additive functor. The right derived functors come from cohomology.

Take a module M and an injective resolution $(Q^\bullet, \eta)$ of M

$$0 \longrightarrow M \xrightarrow{\eta} Q^0 \xrightarrow{d^0} Q^1 \longrightarrow \cdots$$

with each Q^i injective. Then applying F we get

$$0 \longrightarrow F(M) \xrightarrow{F(\eta)} F(Q^0) \xrightarrow{F(d^0)} F(Q^1) \longrightarrow \cdots$$

Considering the sequence

$$0 \longrightarrow F(Q^0) \xrightarrow{F(d^0)} F(Q^1) \xrightarrow{F(d^1)} F(Q^2) \longrightarrow \cdots$$

and $R^nF(M) := H^n(F(Q^\bullet))$. In particular, $R^0F(M) = \ker F(d^0)$.

If $\lambda \in \hom_R(M', M)$, then we can lift λ to $\Lambda : (Q^{\bullet'}, \eta') \to (Q^\bullet, \eta)$ injective resoltions of M' and M respectively. Then $R^nF(\lambda) = \widetilde{F(\Lambda^n)} : R^nF(M') \to R^nF(M)$.

Definition 37. R^nF so defined are the *right derived functors* for F. They are covariant and additive.

There is also a long exact sequence of right derived functors

$$\cdots \to R^n(F(M')) \to R^nF(M) \to R^nF(M'') \to R^{n+1}F(M') \to R^{n+1}F(M) \to \cdots$$

associated to a short exact sequence

$$0 \to M' \to M \to M'' \to 0$$

We get this by taking compatible injective resolutions

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & M' \arrow[r, "\alpha"] \arrow[d, "\eta'"] & M \arrow[r, "\beta"] \arrow[d, "\eta"] & M'' \arrow[r] \arrow[d, "\eta''"] & 0 \\
0 \arrow[r] & Q^{\bullet'} \arrow[r] & Q^\bullet \arrow[r] & Q^{\bullet''} \arrow[r] & 0
\end{tikzcd}$$

and use the Snake Lemma to get our connecting homomorphisms; as the right derived functors are cohomology functors, this induces maps on the chomology groups defining R^nF.

3.12.2 Derived Functors for Contravariant Functors

> "This is just like eating crawfish... You've ever eaten crawfish? You pull off the heads"
>
> Cogdell

Let $G : R - \underline{mod} \to \mathbb{Z} - \underline{mod}$ contravariant and additive functors. We will now construct the left and right derived functors.

Left derived functors still correspond to homology, and the right derived functors correspond to cohomology.

For left derived functors, begin with a module M and an injective resolution $(Q^\bullet, \eta)$

$$0 \to M \to Q^0 \to Q^1 \to Q^2 \to \cdots$$

applying G yields

$$\cdots \to G(Q^2) \to G(Q^1) \to G(Q^0) \to G(M) \to 0$$

considering the complex

$$\cdots \to G(Q^2) \to G(Q^1) \to G(Q^0) \to 0$$

and taking homology, we define $L_nG(M) = H_n(G(Q^\bullet)$. Thus $L_0G(M) = \operatorname{coker} G(d^0)$.

On a morphism $\lambda : M' \to M$ we lift λ to $\Lambda : (Q^{\bullet'}, \eta') \to (Q^\bullet, \eta)$. Applying G and taking homology we get maps on homology $L_nG(\lambda) : L_nG(M) \to L_nG(M')$.

Definition 38. L_nG so defined are the left derived functors of G. They are contravariant and additive.

There are connecting homomorphisms

$$\Delta_n : L_nG(M') \to L_{n-1}G(M'')$$

We get these by taking an exact sequence of R-modules

$$0 \to M' \to M \to M'' \to 0$$

then taking injective resolutions

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & M' \arrow[r, "\alpha"] \arrow[d, "\eta'"] & M \arrow[r, "\beta"] \arrow[d, "\eta"] & M'' \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & Q^{\bullet'} \arrow[r] & Q^\bullet \arrow[r] & Q^{\bullet''} \arrow[r] & 0
\end{tikzcd}$$

and apply G and take homology. With these connecting homomorphisms we get the following long exact sequence

$$\cdots \to L_nG(M'') \to L_nG(M) \to L_nG(M') \to L_{n-1}G(M'') \to L_{n-1}G(M) \to \cdots$$

Similarly for the right derived functors of G we begin with M and a projective resolution $(P_\bullet, \epsilon)$

$$\cdots \to P_2 \to P_1 \to P_0 \to M \to 0$$

apply G to get

$$0 \to G(M) \to G(P_0) \to G(P_1) \to G(P_2) \to \cdots$$

reduce to

$$0 \to G(P_0) \to G(P_1) \to G(P_2) \to \cdots$$

taking cohomology

$$R^nG(M) = H^n(G(P_\bullet))$$

and $R^0G(M) = \ker G(d^1)$.

If $\mu : M' \to M$ is a morphism, we lift μ to a morphism of resolution $(P'_\bullet, \epsilon') \to (P_\bullet, \epsilon)$ then apply G, then take out $G(M), G(M')$ in the sequences and get the induced maps on cohomology. This gives up maps $R^nG(\mu) : R^nG(M) \to R^nG(M')$.

Definition 39. The $R^nG(M)$ so defined are the *right derived functors* for G. They are contravariant, additive functors.

In a similar way we get a long exact sequence of right derived functors

$$R^nG(M'') \to R^nG(M) \to R^nG(M') \to R^{n+1}G(M'') \to R^{n+1}G(M) \to \cdots$$

from a short exact sequence of R-modules

$$0 \to M' \to M \to M'' \to 0$$